

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/notswitched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/notswitched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4

```

```
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)
```

```
return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168

Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping : id_channel

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

Grouping : id_channel

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

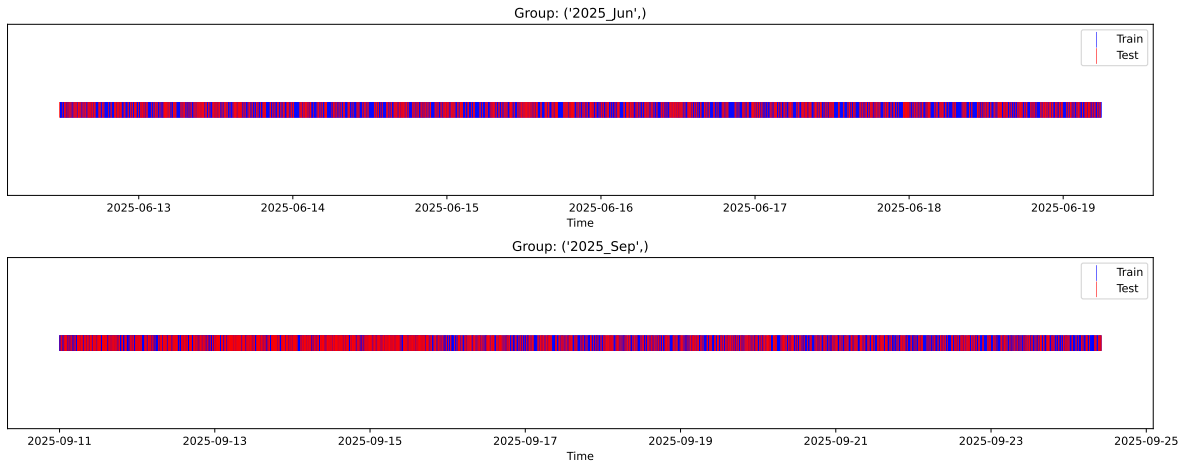
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_{train} vs X_{test}

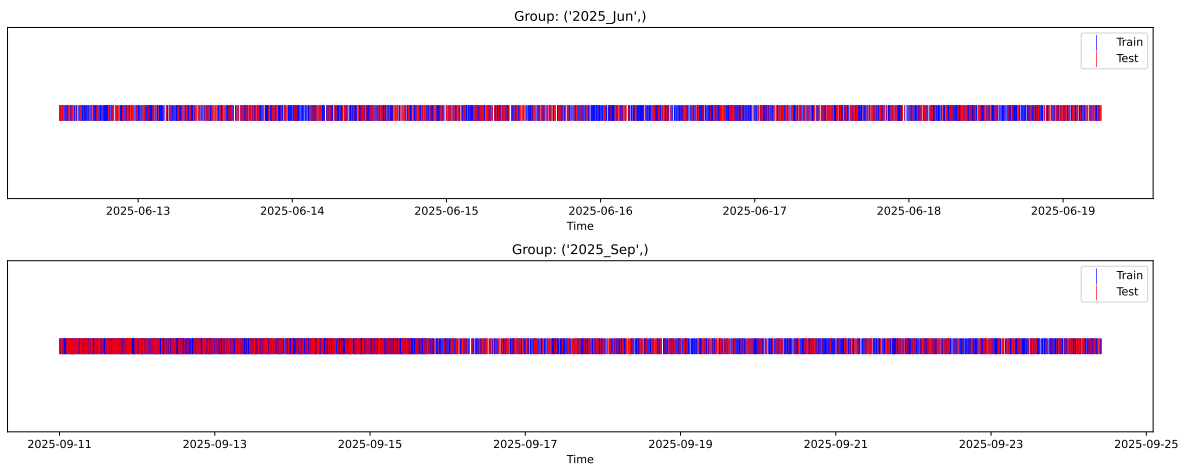
External Split: Train: 5717 samples Test: 2451 samples



2.2.2 Internal CV Folds

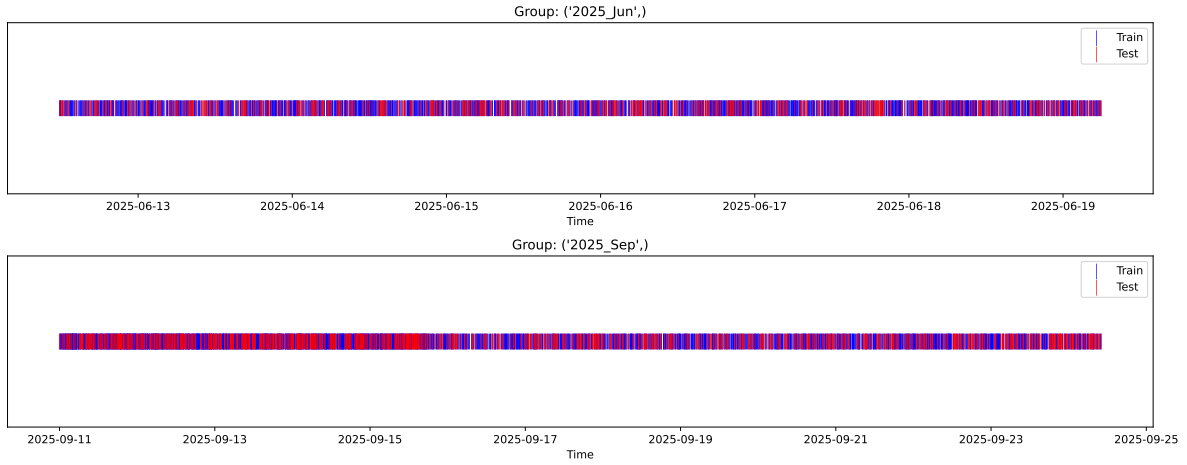
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



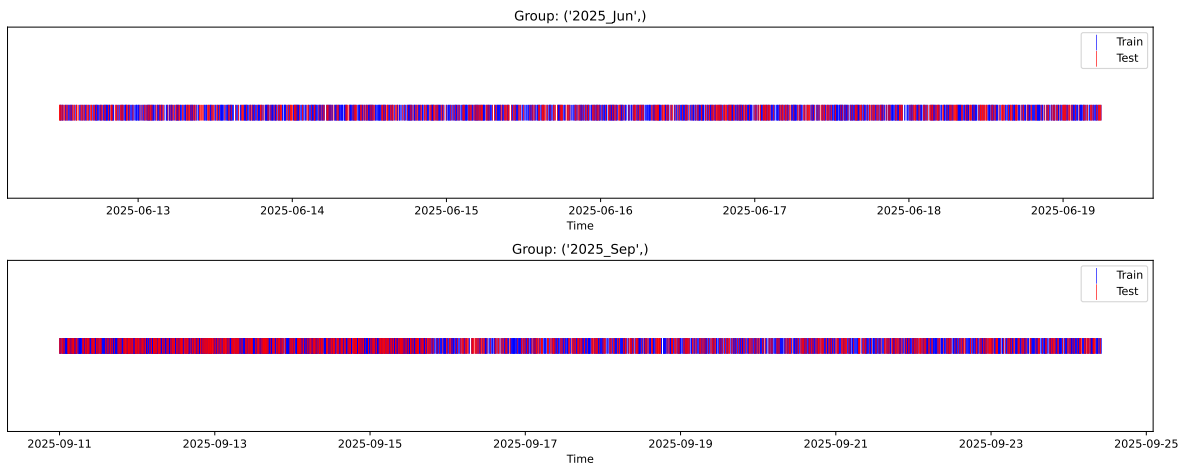
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



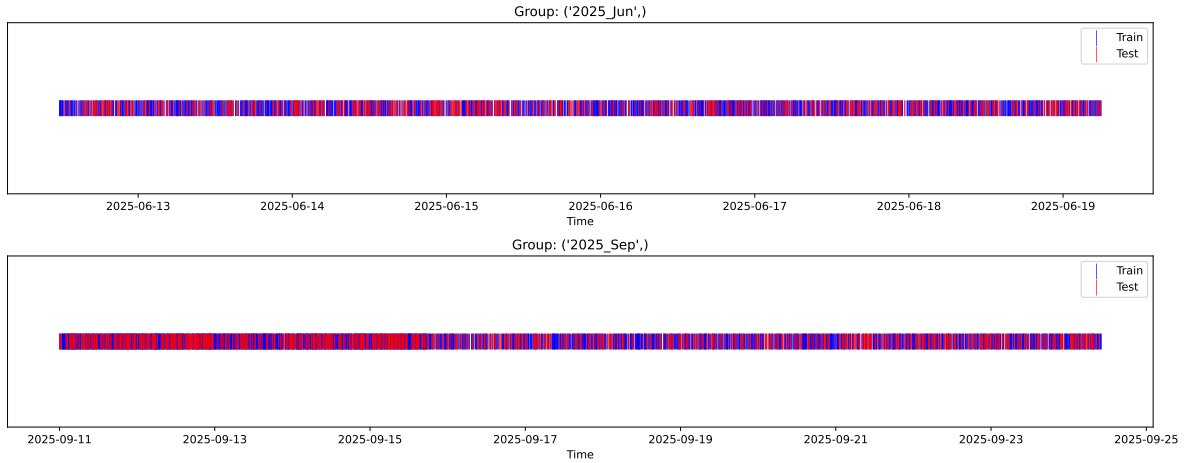
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



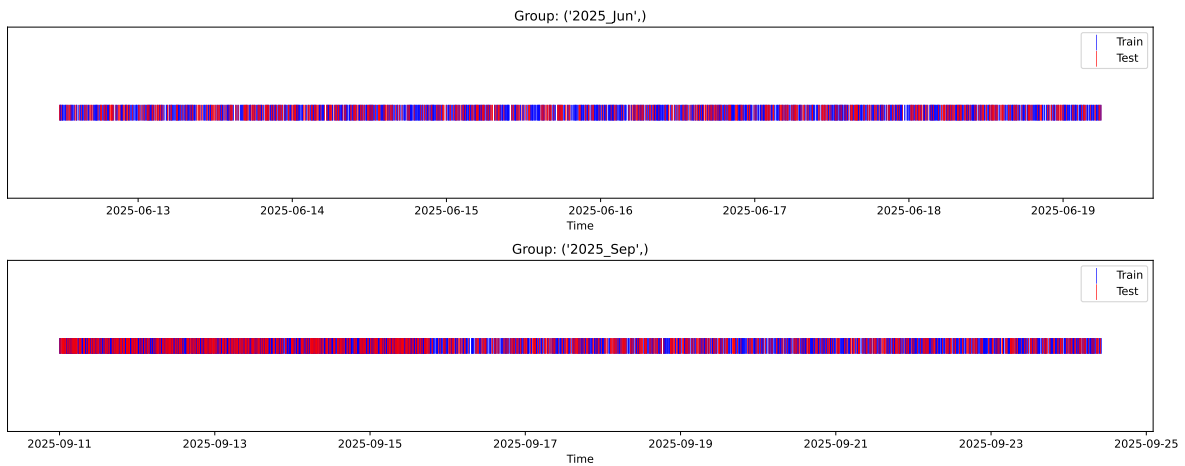
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



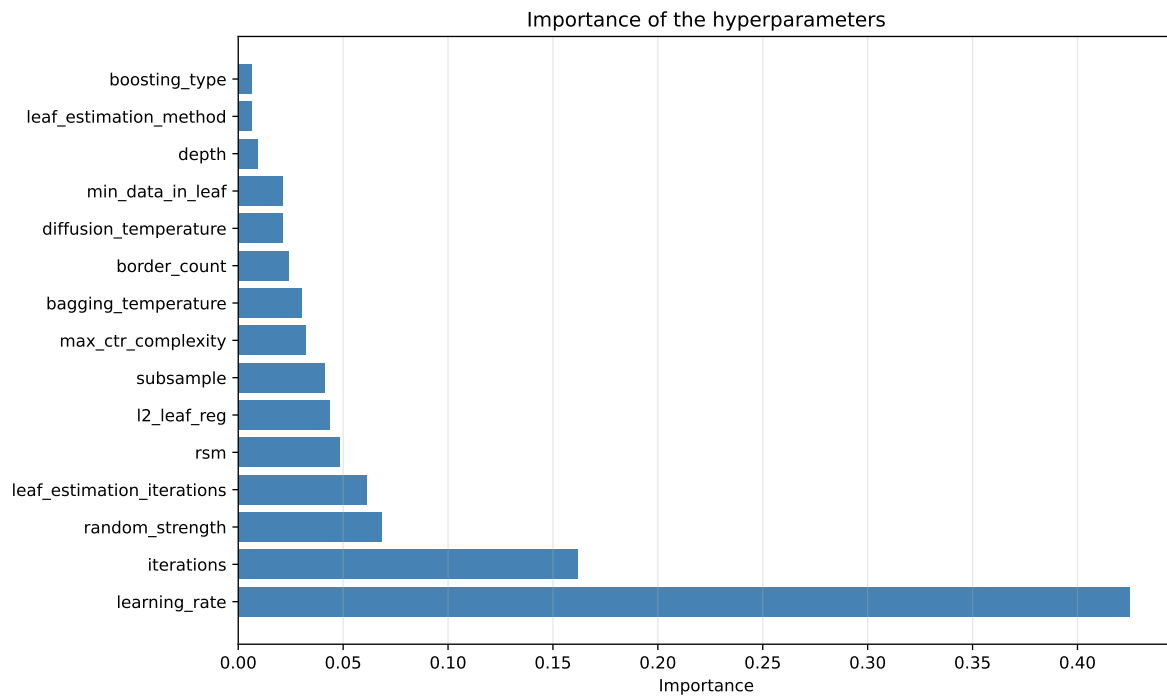
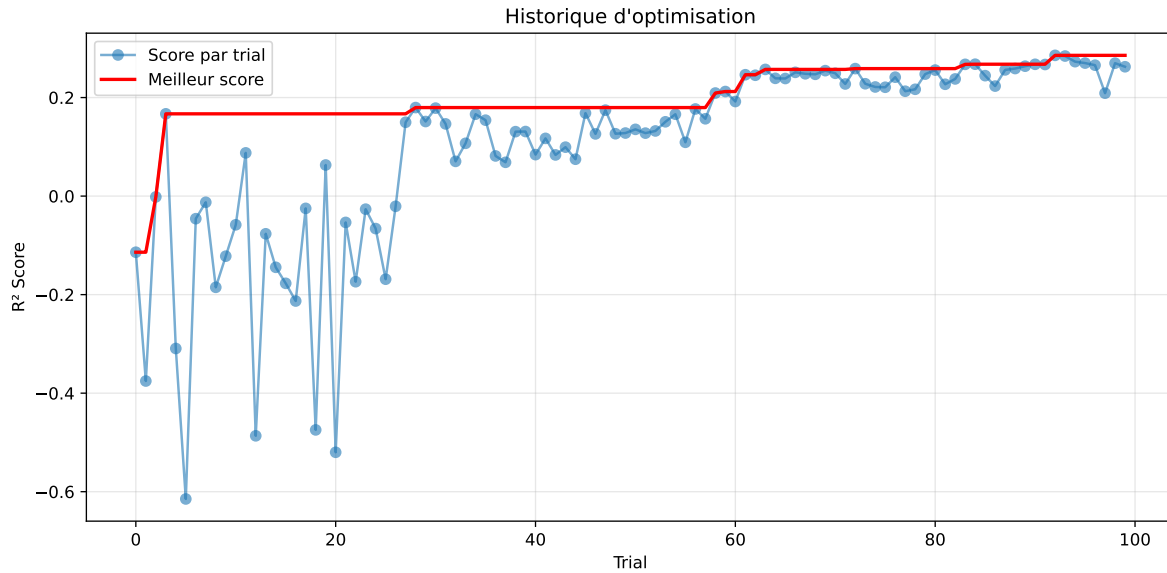
2.3 Model : CatBoost

2.3.1 Gridsearch — Group: 3

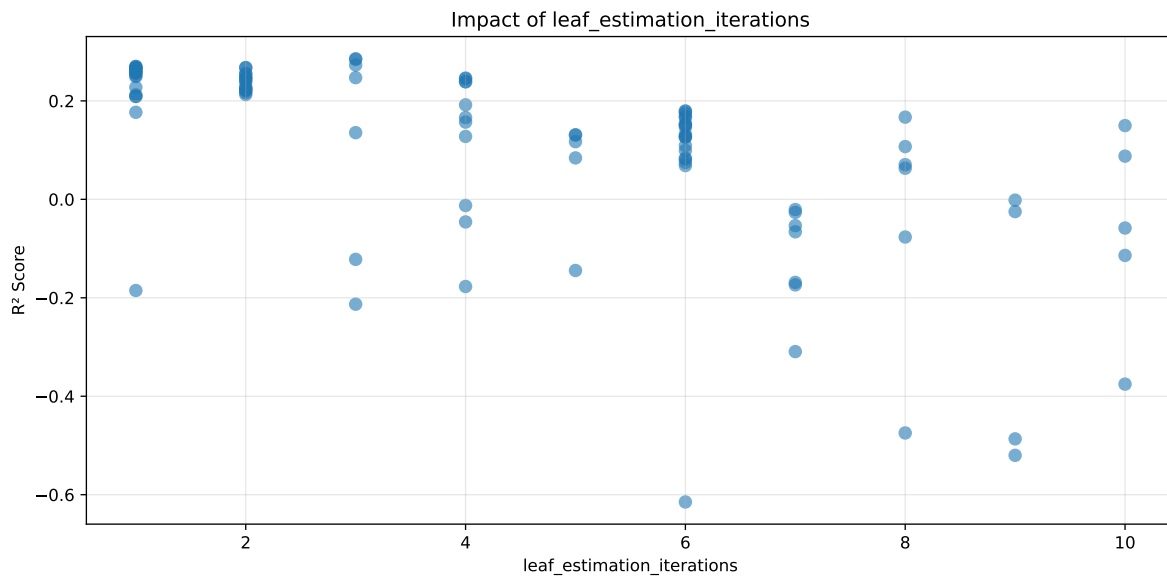
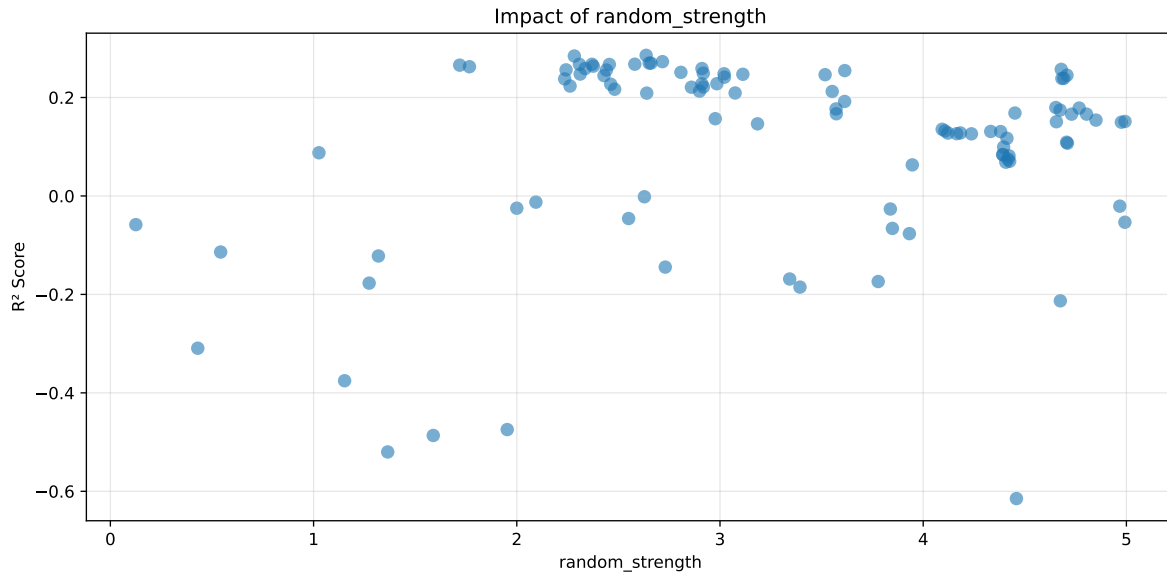
Len(group) : 1356, Len(training) : 950, Len(test) : 406

Distribution y_train vs y_test: y_train: mean=2.317, std=0.708, min=1.041, max=6.086
y_test: mean=2.318, std=0.740, min=1.068, max=5.447

===== TOP Importance FEATURES =====
feature importance r_umidity 67.266810 r_temperature 10.563510 r_NH3 9.189931 r_THI
6.748201 r_CO2 6.231548

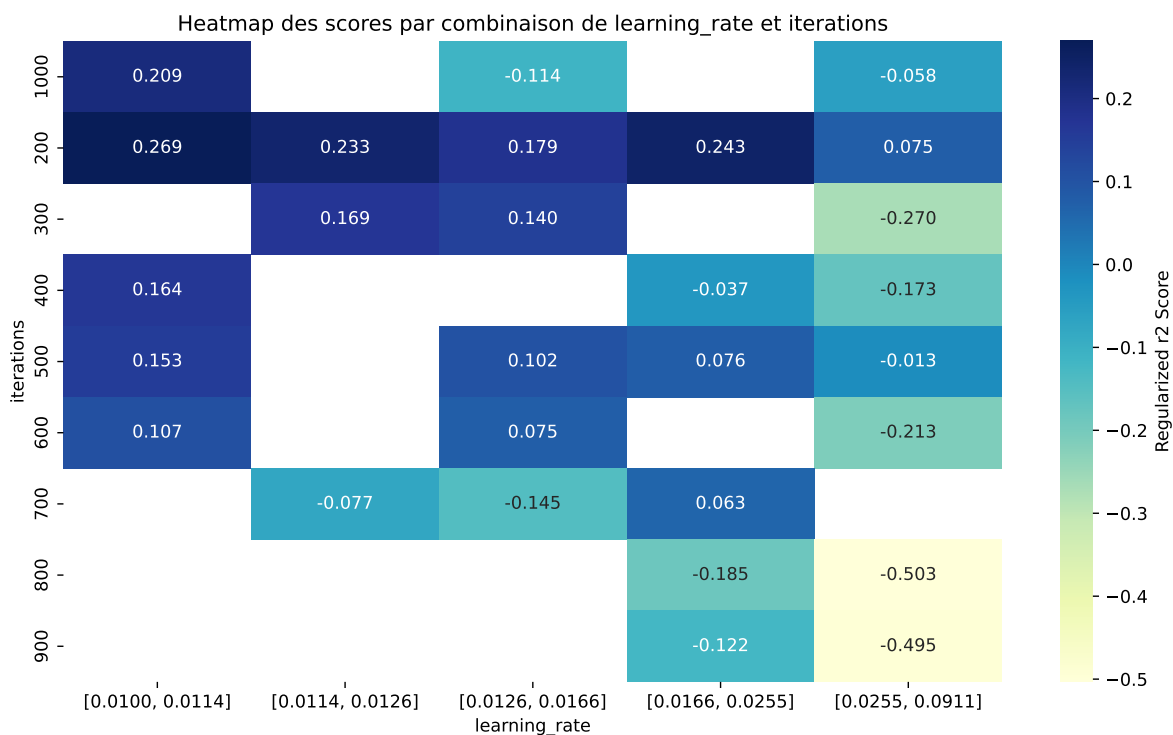


Paramètres avec >5% d'importance : ['learning_rate', 'iterations', 'random_strength', 'leaf_estimation_iterations']



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The default value of `observed=False` is deprecated and will change to `observed=True` in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6221	0.566	0.6371	0.0388	0.2857	0.6461	0.0801	1.1415	1.0991
2	0.625	0.5682	0.6403	0.0396	0.2843	0.6464	0.0782	1.1377	1.0999
3	0.6223	0.5641	0.635	0.0373	0.2729	0.6414	0.0773	1.1371	1.1033
4	0.5827	0.5306	0.5977	0.0367	0.27	0.6139	0.0833	1.157	1.0982
5	0.5682	0.5185	0.5861	0.036	0.2698	0.6017	0.0832	1.1605	1.0959
6	0.6155	0.5576	0.6308	0.0366	0.2677	0.6382	0.0807	1.1447	1.104
7	0.6544	0.5899	0.6719	0.0389	0.2675	0.6814	0.0915	1.155	1.1093
8	0.6163	0.5582	0.632	0.0365	0.2674	0.6398	0.0816	1.1462	1.1042
9	0.6566	0.5917	0.6723	0.0376	0.2671	0.6817	0.0901	1.1522	1.1097
10	0.5834	0.5305	0.6033	0.0373	0.2657	0.6151	0.0846	1.1594	1.0998
---	---	---	---	---	---	---	---	---	---
100	0.9563	0.6945	0.7407	0.014	-0.6148	0.7605	0.0661	1.0951	1.377

Hyperparameters:

Rank 1: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.011117660823257845, 'reg__l2_leaf_reg': 35.58491434775047, 'reg__bagging_temperature': 4.58716832061366, 'reg__random_strength': 2.6366475570190686, 'reg__subsample': 0.6339077409317998, 'reg__min_data_in_leaf': 29, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.6256663970788574, 'reg__border_count': 40, 'reg__leaf_estimation_method': 'Gradient',

'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 5625.941539543665}

Rank 2: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.01111445365262063, 'reg__l2_leaf_reg': 35.464291977533335, 'reg__bagging_temperature': 4.584950916567565, 'reg__random_strength': 2.283229526196379, 'reg__subsample': 0.6378679751067892, 'reg__min_data_in_leaf': 32, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.6262075448187153, 'reg__border_count': 35, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 6574.750978040078}

Rank 3: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.01109845613391184, 'reg__l2_leaf_reg': 35.359758967474185, 'reg__bagging_temperature': 4.681247808100292, 'reg__random_strength': 2.716798862063589, 'reg__subsample': 0.773090674051025, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.628980003689174, 'reg__border_count': 65, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 4975.198270489141}

Rank 4: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.011188616963830332, 'reg__l2_leaf_reg': 35.416850406140036, 'reg__bagging_temperature': 4.610809060436626, 'reg__random_strength': 2.6514150567300847, 'reg__subsample': 0.7879839281133528, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6322431787021591, 'reg__border_count': 70, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 4844.1855942032535}

Rank 5: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.010685649723350614, 'reg__l2_leaf_reg': 37.04335135338566, 'reg__bagging_temperature': 4.969585522526076, 'reg__random_strength': 2.6604420876578656, 'reg__subsample': 0.768329420913072, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6422673915997696, 'reg__border_count': 66, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 4867.78957494624}

Rank 6: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.01098216066872142, 'reg__l2_leaf_reg': 34.769266369612424, 'reg__bagging_temperature': 4.507467435040699, 'reg__random_strength': 2.5806714300440103, 'reg__subsample': 0.7631642008313024, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6632601458774312, 'reg__border_count': 83, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 5833.7722767708265}

Rank 7: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.02327859793789078, 'reg__l2_leaf_reg': 35.86249887777121, 'reg__bagging_temperature': 4.6323211740133505, 'reg__random_strength': 2.3701275696311868, 'reg__subsample': 0.7549409077056553,

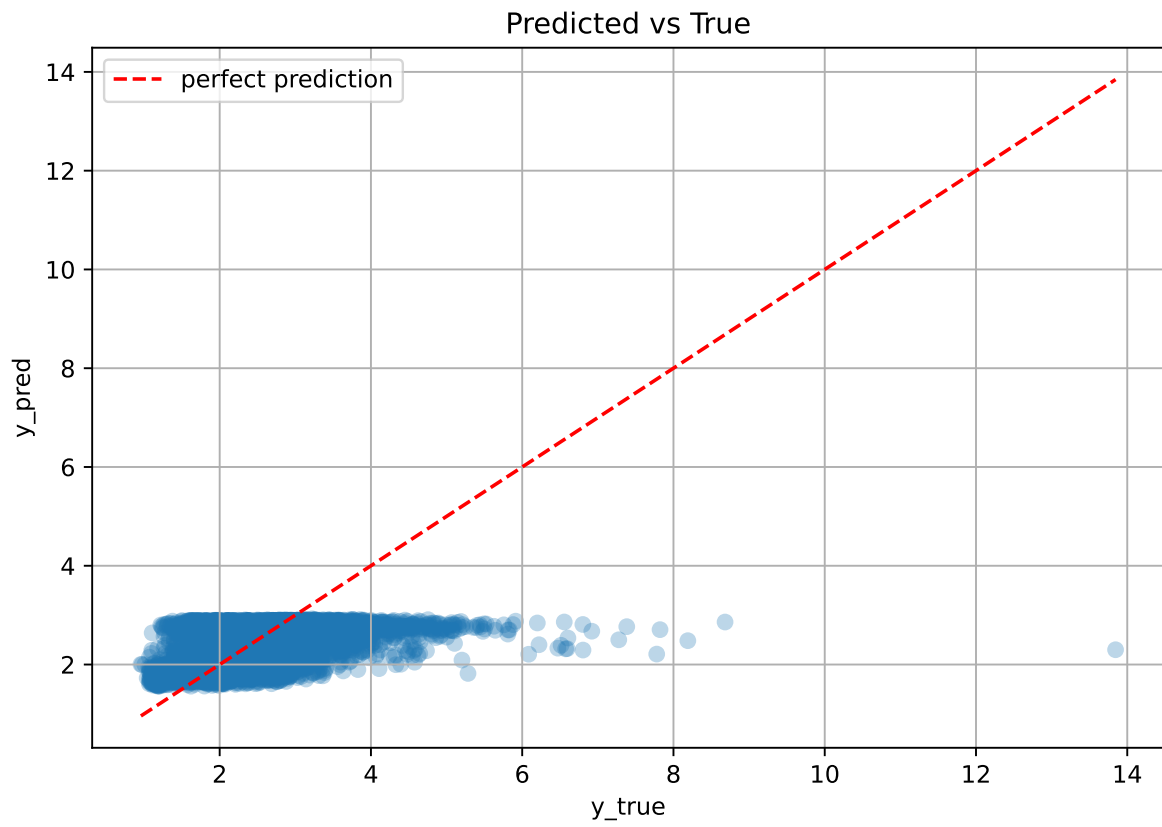
'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6337678832510012, 'reg__border_count': 64, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 6413.852833815098}

Rank 8: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.011060321922237751, 'reg__l2_leaf_reg': 35.05286808997433, 'reg__bagging_temperature': 4.266071785574594, 'reg__random_strength': 2.3082465807736035, 'reg__subsample': 0.7442583941242291, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6617272982080812, 'reg__border_count': 83, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 5729.9996295577375}

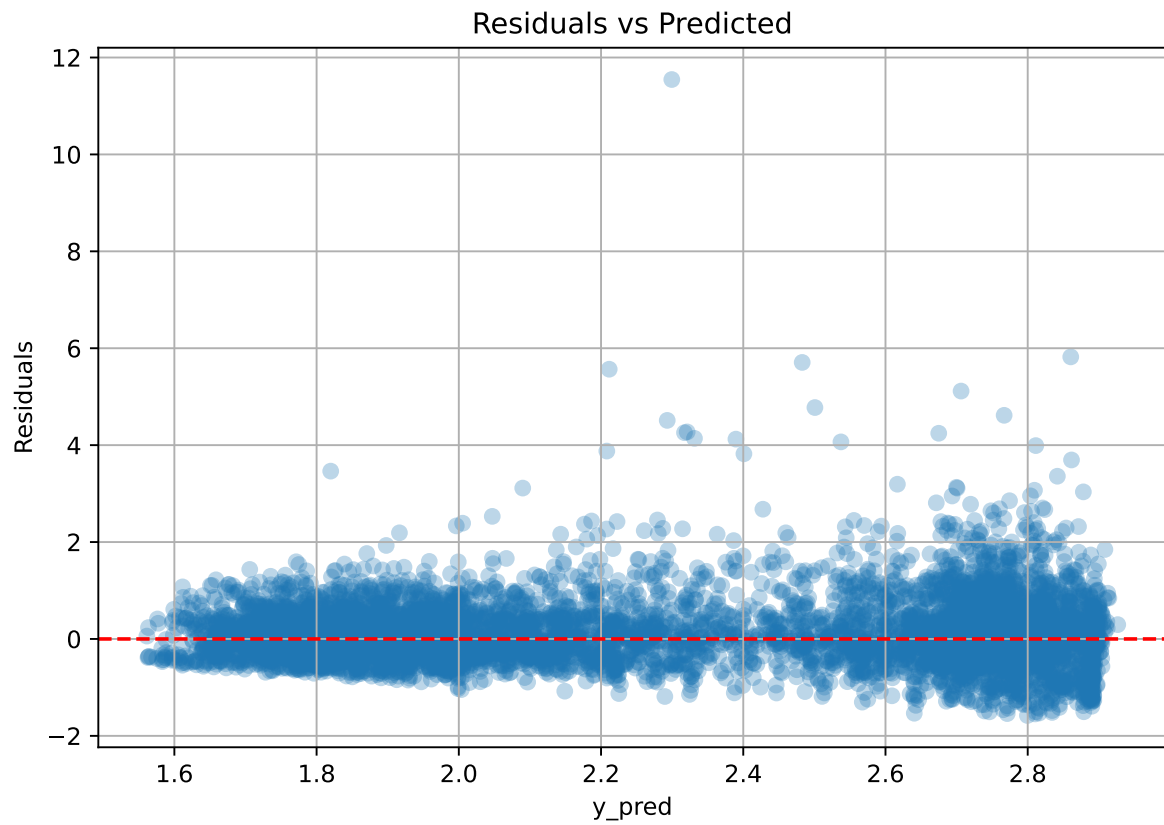
Rank 9: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.023112982874078602, 'reg__l2_leaf_reg': 35.24523298947062, 'reg__bagging_temperature': 4.490104927599475, 'reg__random_strength': 2.455440292775438, 'reg__subsample': 0.7309061480516207, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6434547583398091, 'reg__border_count': 64, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 5751.46566005583}

Rank 10: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.011117466647038818, 'reg__l2_leaf_reg': 36.66113255683489, 'reg__bagging_temperature': 4.485349890680351, 'reg__random_strength': 1.7190378613430162, 'reg__subsample': 0.7861721845233551, 'reg__min_data_in_leaf': 29, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.627966150963648, 'reg__border_count': 87, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 6715.195616580906}

2.3.2 Scatter



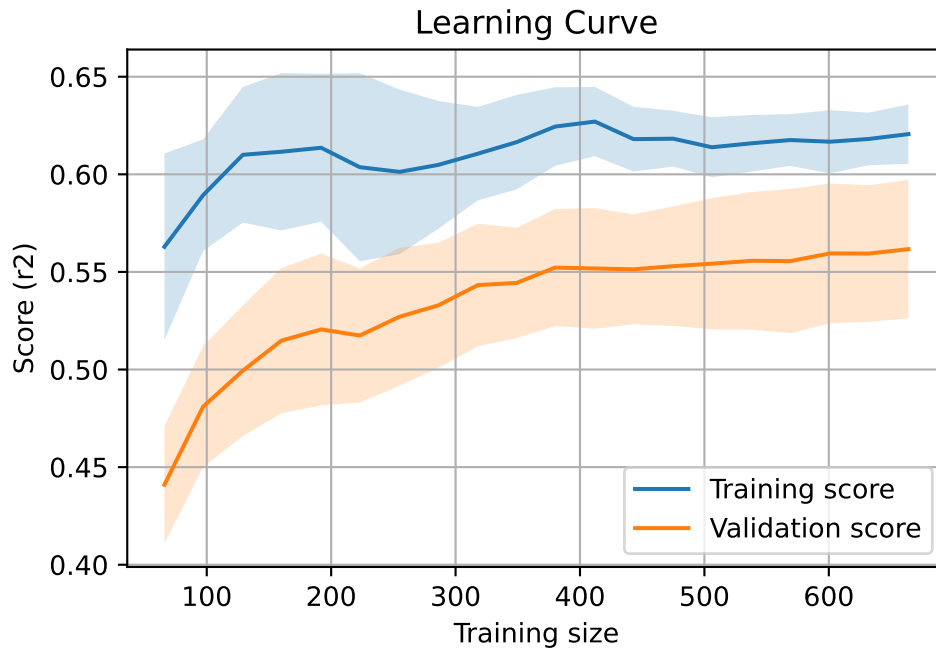
2.3.3 Scatter residuals



2.3.4 Learning curve — Group: 3

`Len(group)` : 1356, `Len(training)` : 950,

`Len(training_real)` : 665, `Len(test_of_training)` : 285, `Len(test)` : 406

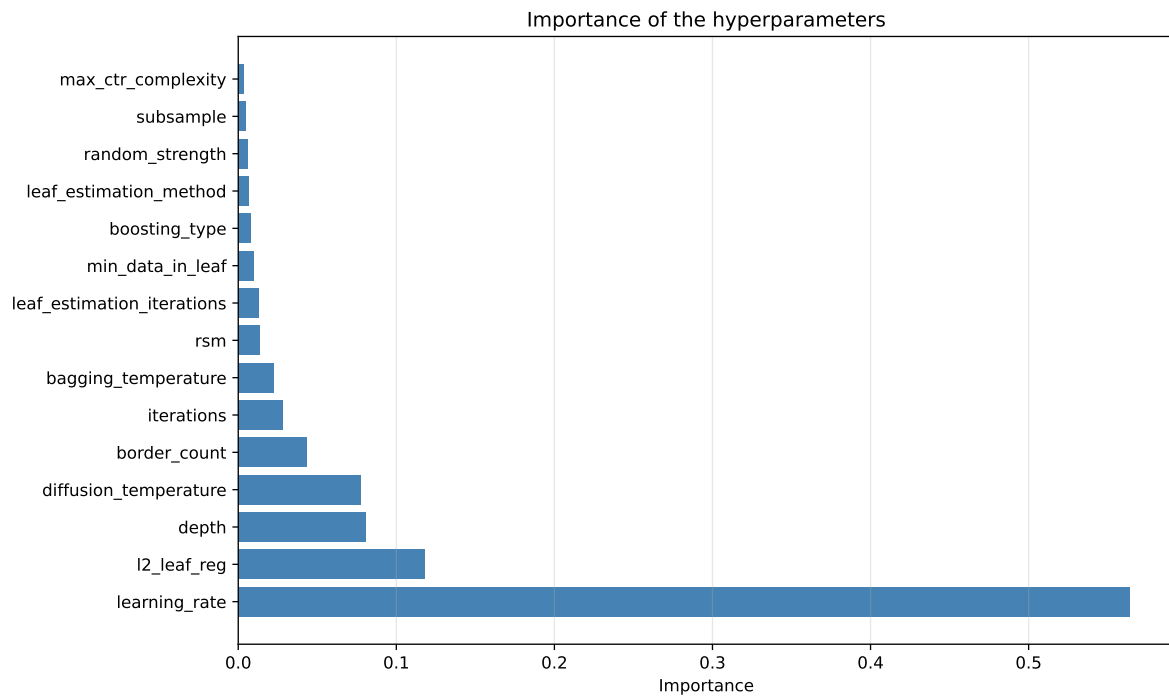
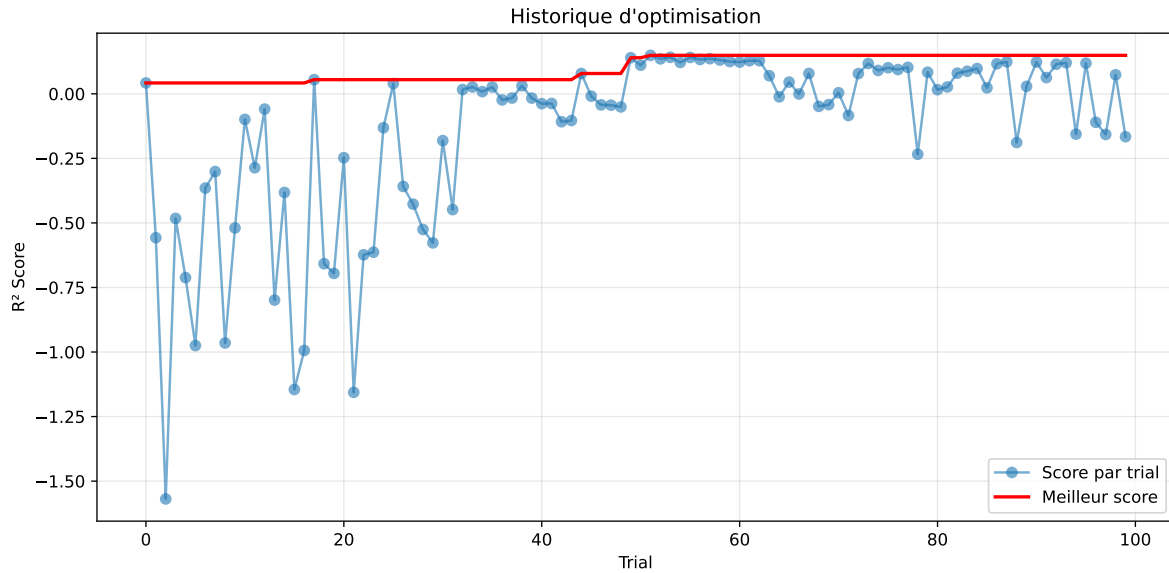


2.3.5 Gridsearch — Group: 4

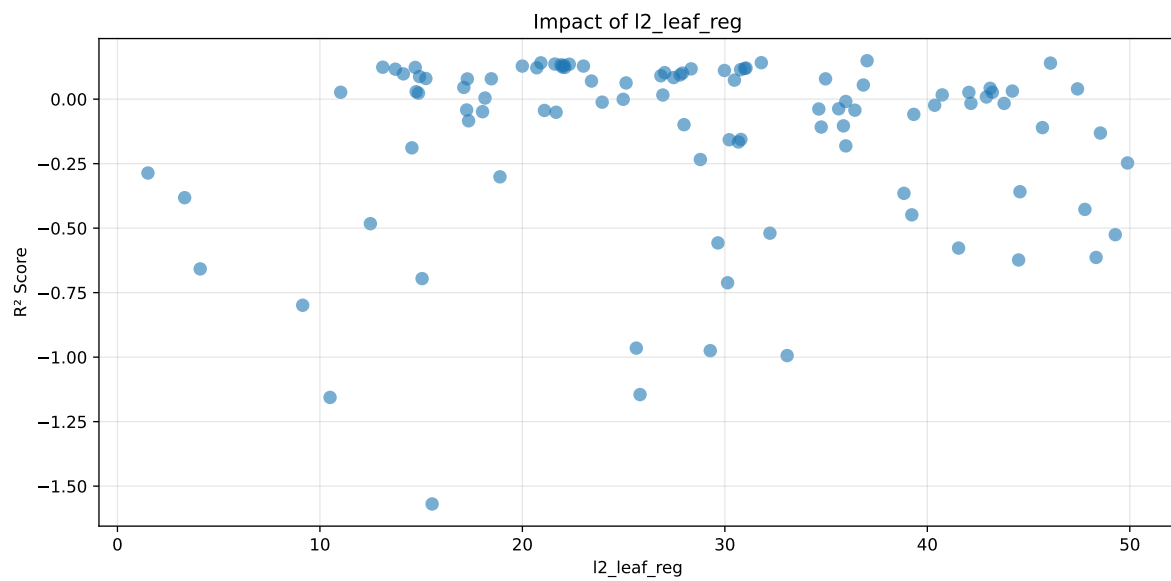
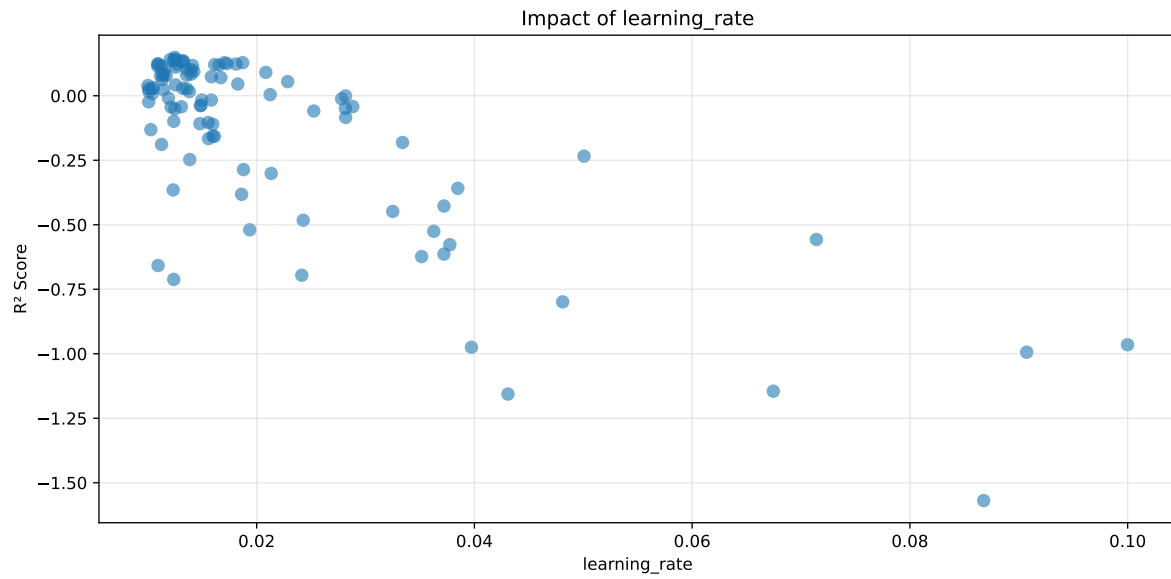
Len(group) : 1360, Len(training) : 952, Len(test) : 408

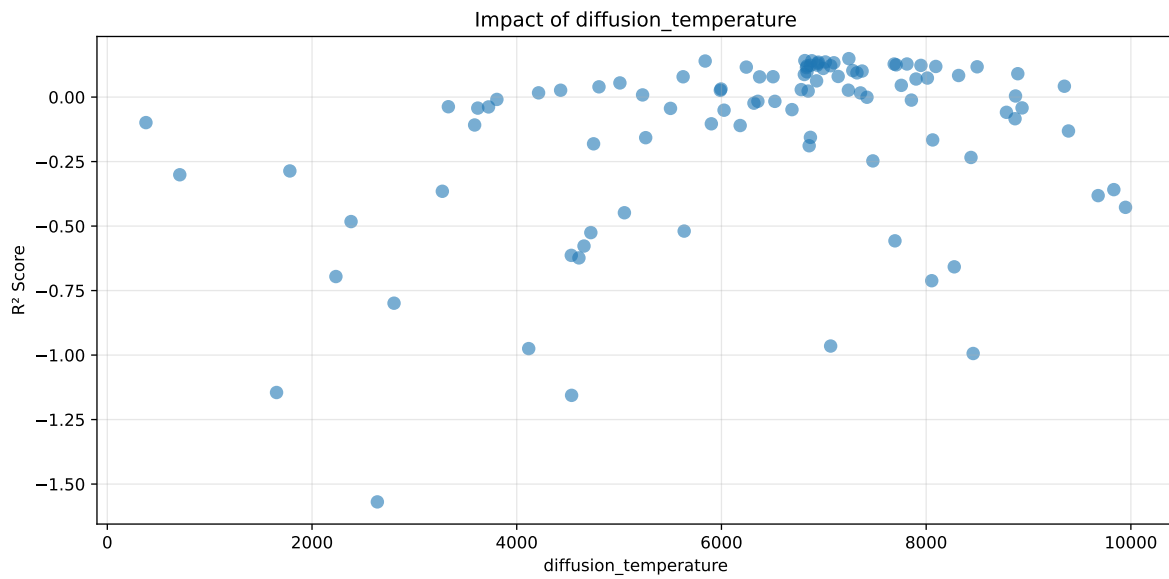
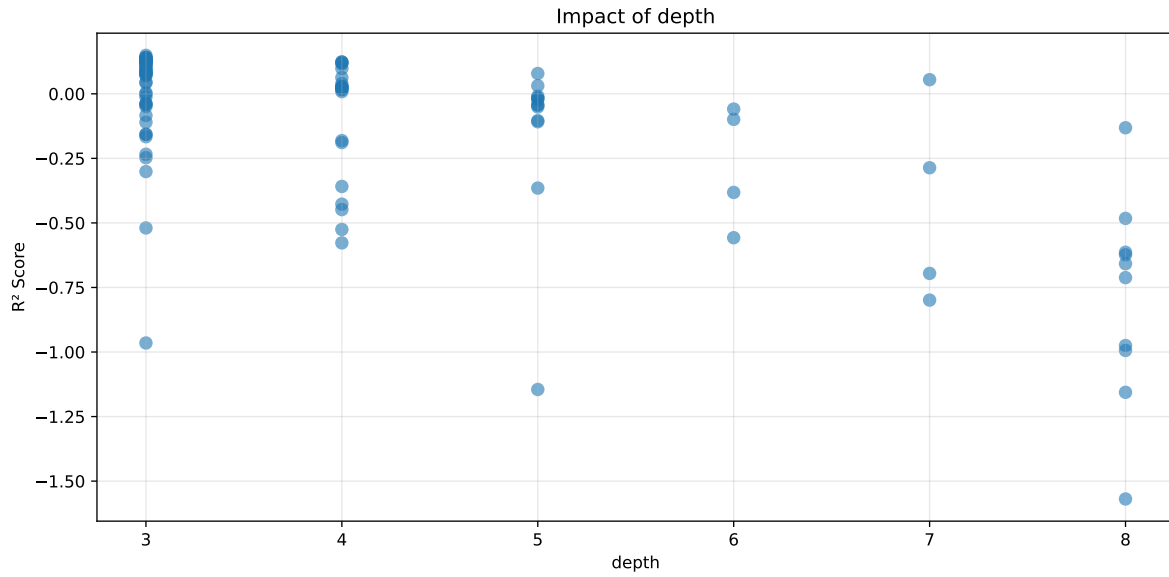
Distribution y_train vs y_test: y_train: mean=2.239, std=0.831, min=0.959, max=7.383
y_test: mean=2.202, std=0.791, min=1.062, max=7.822

===== TOP Importance FEATURES =====
feature importance r_umidity 52.820669 r_THI 24.451535 r_CO2 9.717165 r_NH3 7.522299
r_temperature 5.488333



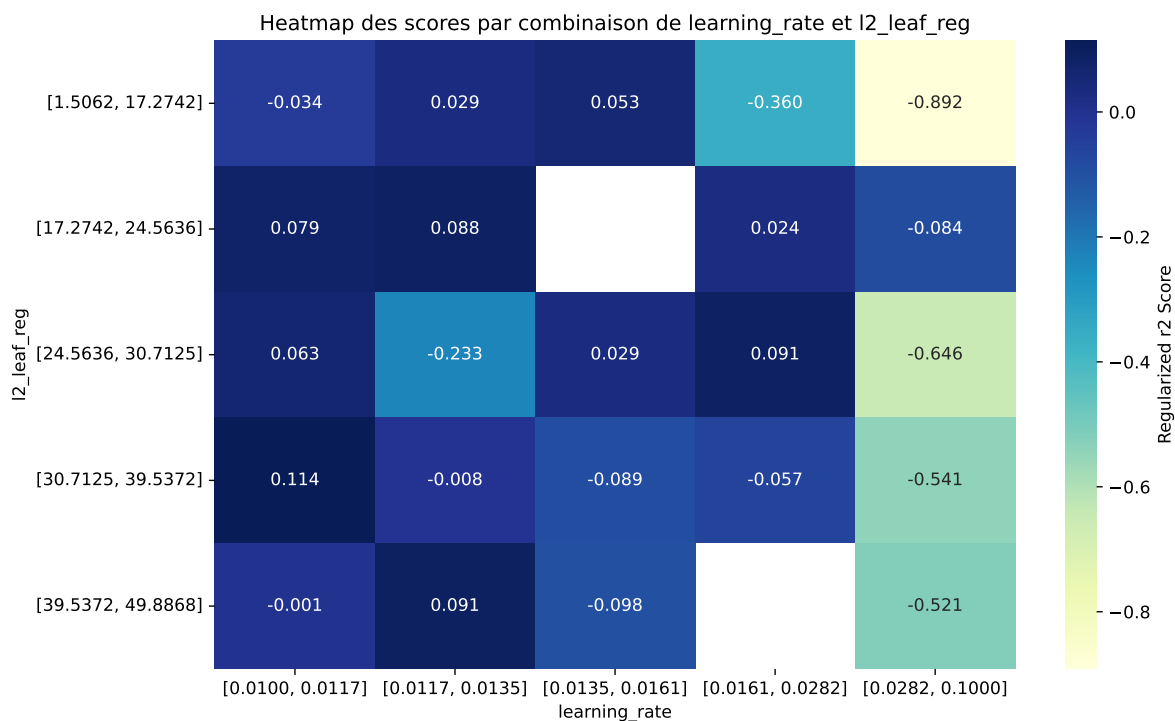
Paramètres avec >5% d'importance : ['learning_rate', 'l2_leaf_reg', 'depth', 'diffusion_temperature']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.3419	0.3098	0.3324	0.0393	0.149	0.341	0.0312	1.1007	1.1038
2	0.3496	0.3149	0.3419	0.0397	0.1414	0.3457	0.0308	1.0979	1.1102
3	0.362	0.3251	0.3524	0.0395	0.1407	0.3548	0.0297	1.0915	1.1134
4	0.3506	0.3154	0.3467	0.0398	0.1397	0.3515	0.0361	1.1143	1.1114
5	0.3731	0.3336	0.3619	0.0416	0.1361	0.3719	0.0382	1.1146	1.1184
6	0.3724	0.3329	0.3541	0.0405	0.1354	0.3657	0.0328	1.0985	1.1186
7	0.3748	0.3345	0.3641	0.0378	0.1329	0.3766	0.0421	1.1258	1.1205
8	0.3748	0.3341	0.3654	0.0399	0.1309	0.3767	0.0426	1.1275	1.1216
9	0.4297	0.3794	0.4157	0.0432	0.1283	0.4211	0.0416	1.1098	1.1324
10	0.4253	0.3758	0.4108	0.0432	0.1281	0.4121	0.0363	1.0966	1.1318
---	---	---	---	---	---	---	---	---	---
100	0.9939	0.5667	0.5987	0.0371	-1.5691	0.6461	0.0793	1.14	1.7537

Hyperparameters:

Rank 1: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.012472336611392035, 'reg__l2_leaf_reg': 37.02418724365619, 'reg__bagging_temperature': 4.559671360041633, 'reg__random_strength': 3.128822194532666, 'reg__subsample': 0.6376416765811085, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.8070692305026868, 'reg__border_count': 45, 'reg__leaf_estimation_method': 'Gradient',

'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 7244.564845791208}

Rank 2: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.012551798968450233, 'reg__l2_leaf_reg': 31.80030235710437, 'reg__bagging_temperature': 4.642391523011801, 'reg__random_strength': 2.929894324449679, 'reg__subsample': 0.6586749146114644, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.8935421814910053, 'reg__border_count': 47, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 6814.767796461205}

Rank 3: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.012067237323965543, 'reg__l2_leaf_reg': 20.911907139601126, 'reg__bagging_temperature': 4.689780776759764, 'reg__random_strength': 2.844075520470823, 'reg__subsample': 0.6525834428129352, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.8967922919659763, 'reg__border_count': 45, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 6883.056822076407}

Rank 4: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.012457057483323198, 'reg__l2_leaf_reg': 46.081774161644205, 'reg__bagging_temperature': 4.347326695592112, 'reg__random_strength': 4.625301184331387, 'reg__subsample': 0.727037222045863, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.8244850316589045, 'reg__border_count': 48, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5841.044045776196}

Rank 5: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.013239672580573913, 'reg__l2_leaf_reg': 21.602528429191253, 'reg__bagging_temperature': 4.656812532999614, 'reg__random_strength': 2.7617747719434185, 'reg__subsample': 0.6476656565136162, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.8254291658334437, 'reg__border_count': 47, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 7015.002750990547}

Rank 6: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.01236168583793496, 'reg__l2_leaf_reg': 22.3161455088064, 'reg__bagging_temperature': 4.548778964289747, 'reg__random_strength': 2.732465801920042, 'reg__subsample': 0.6467328481259724, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.8210380801668233, 'reg__border_count': 74, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 6946.552683224443}

Rank 7: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.013126465989278108, 'reg__l2_leaf_reg': 21.906802590055825, 'reg__bagging_temperature': 4.653356645635382, 'reg__random_strength': 2.6716568718840272, 'reg__subsample': 0.6503912764017271,

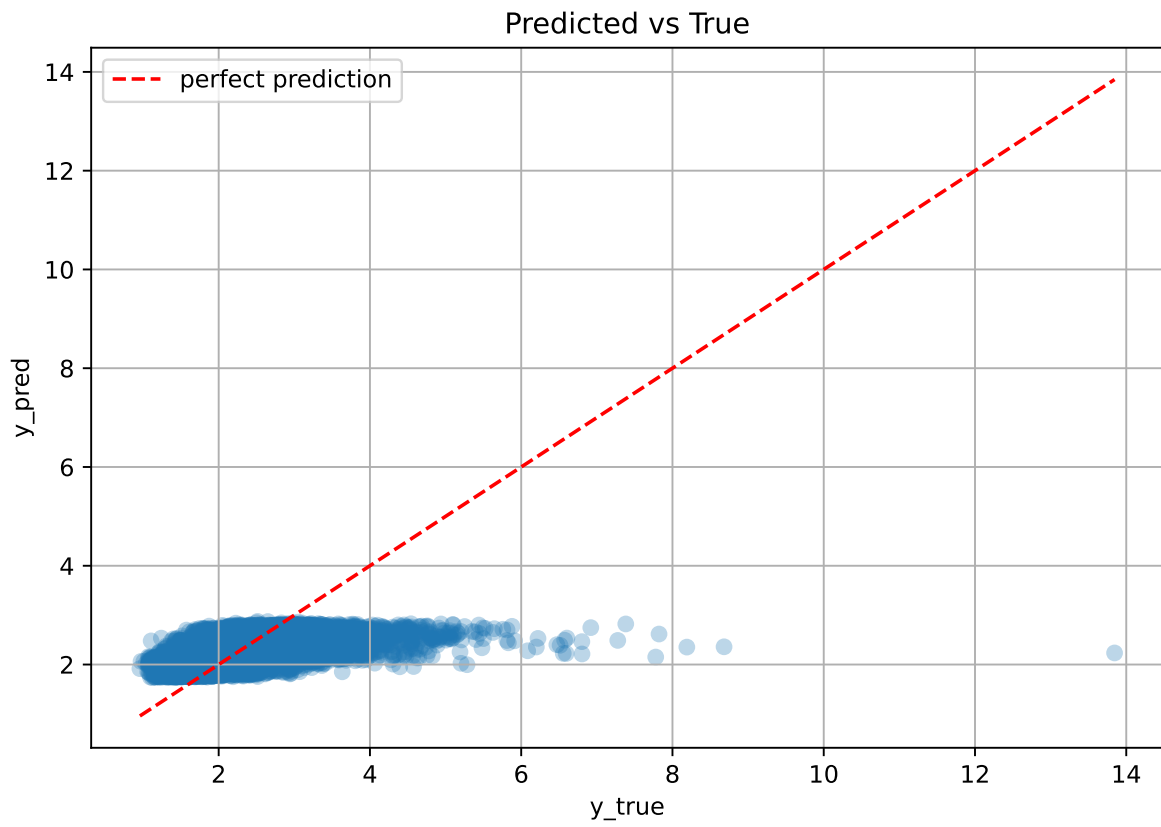
'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.8263316910878177, 'reg__border_count': 48, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 7098.376711918498}

Rank 8: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.01330243697871785, 'reg__l2_leaf_reg': 22.05452531153734, 'reg__bagging_temperature': 4.752432951576186, 'reg__random_strength': 2.7209124846245327, 'reg__subsample': 0.6495990752010944, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.8968523132765249, 'reg__border_count': 48, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 6940.865873993547}

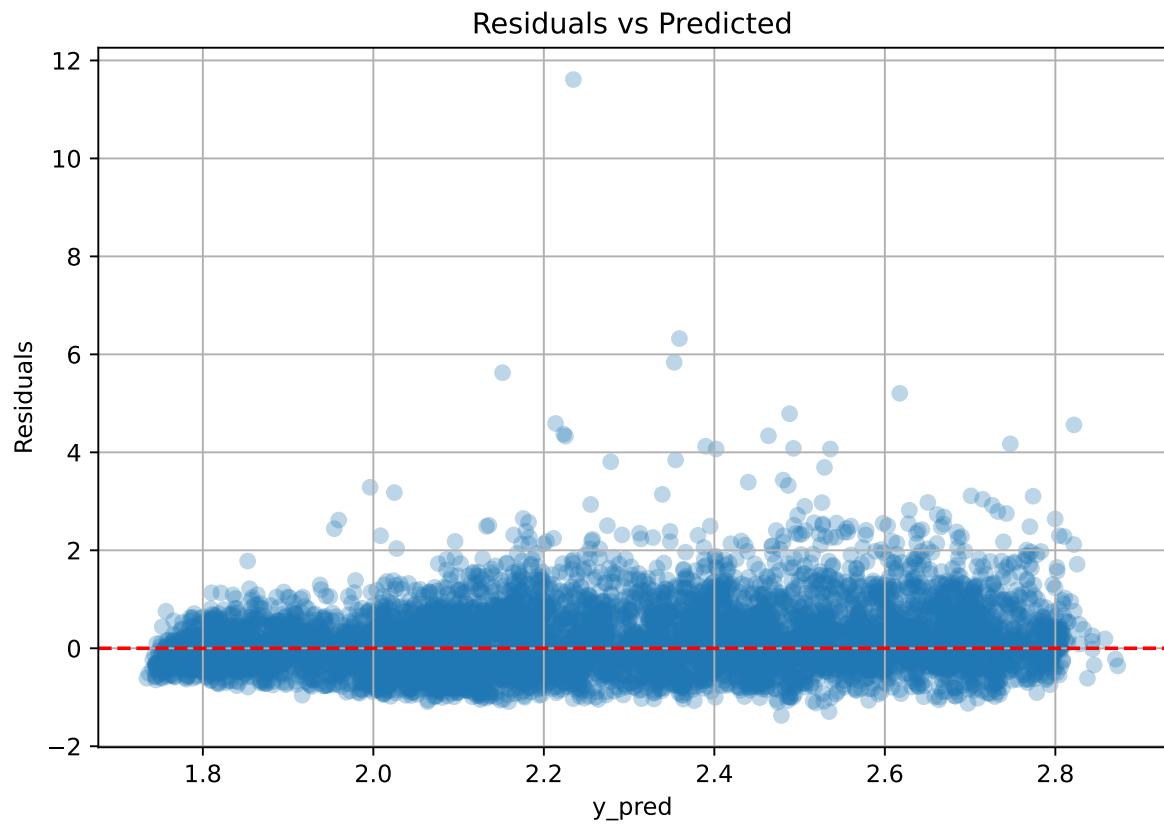
Rank 9: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.018717752492061042, 'reg__l2_leaf_reg': 23.016180410332726, 'reg__bagging_temperature': 4.840515338788277, 'reg__random_strength': 2.7854977633287685, 'reg__subsample': 0.6259904489438282, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.8384722101573263, 'reg__border_count': 46, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 7812.402232796867}

Rank 10: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.017038577048741718, 'reg__l2_leaf_reg': 19.99525728309832, 'reg__bagging_temperature': 4.8467004036921395, 'reg__random_strength': 2.8130936171485743, 'reg__subsample': 0.6400351105871989, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.8391734184725624, 'reg__border_count': 48, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 7687.480798769955}

2.3.6 Scatter



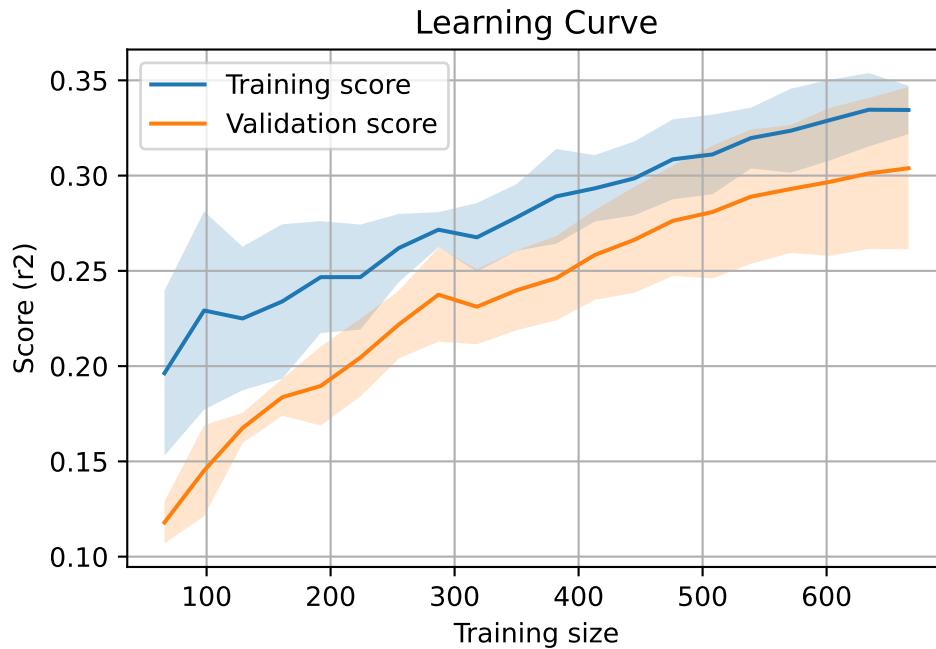
2.3.7 Scatter residuals



2.3.8 Learning curve — Group: 4

`Len(group)` : 1360, `Len(training)` : 952,

`Len(training_real)` : 667, `Len(test_of_training)` : 285, `Len(test)` : 408

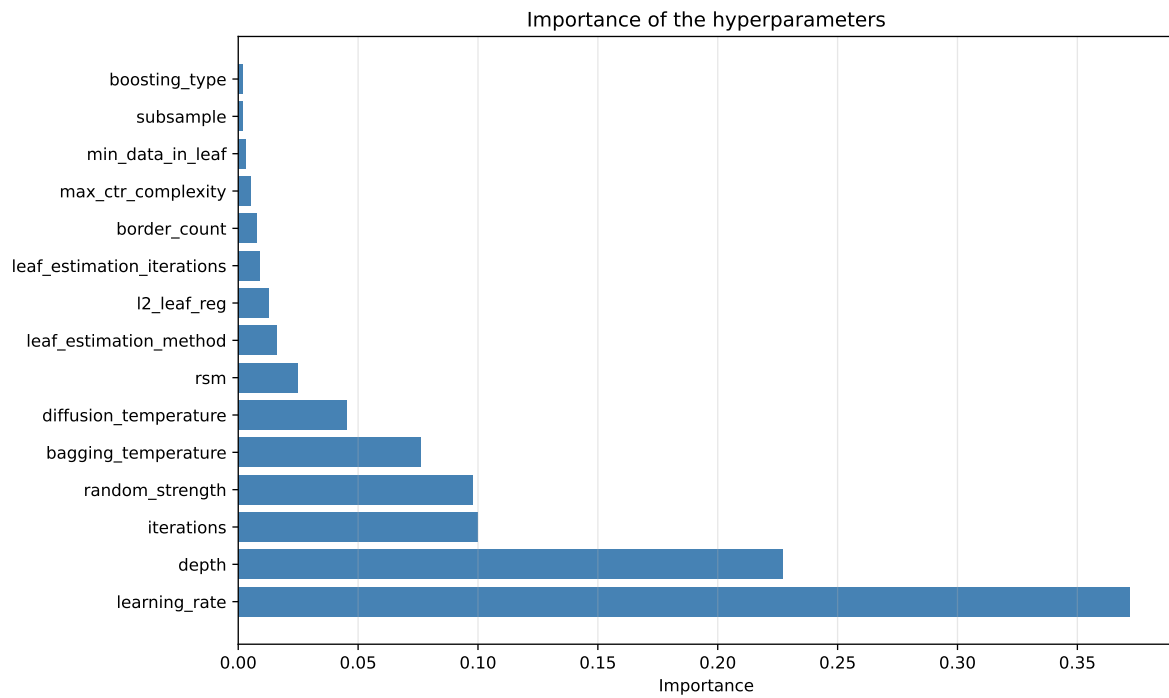
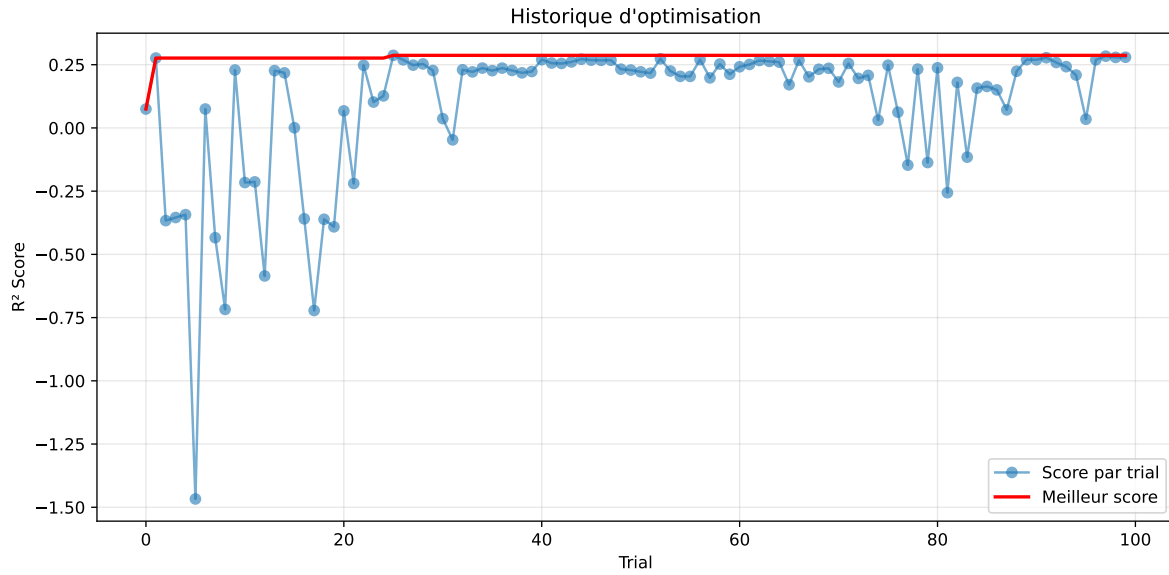


2.3.9 Gridsearch — Group: 5

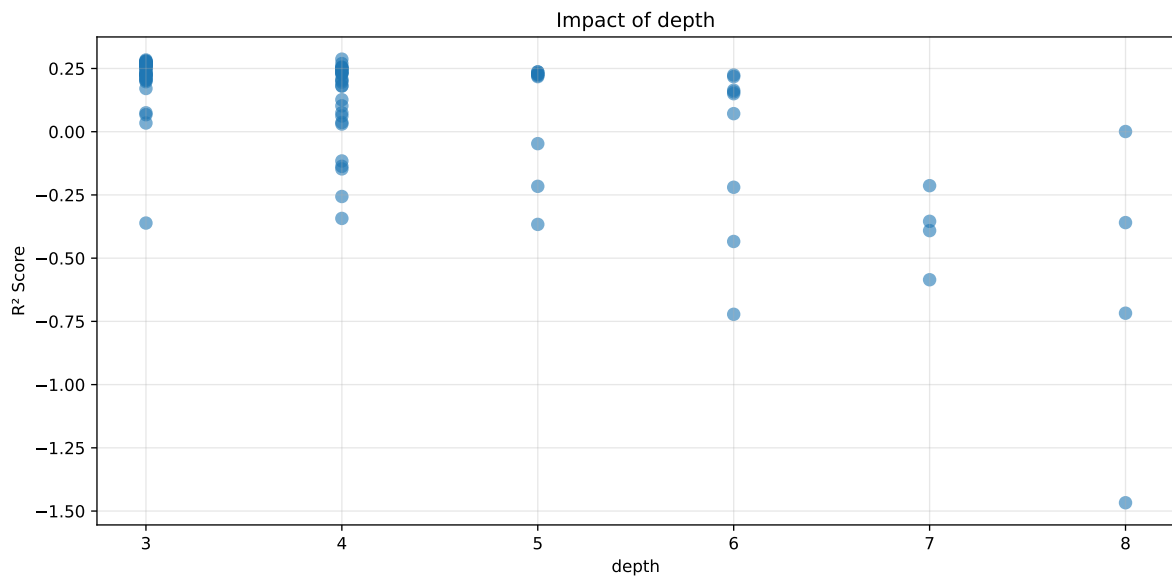
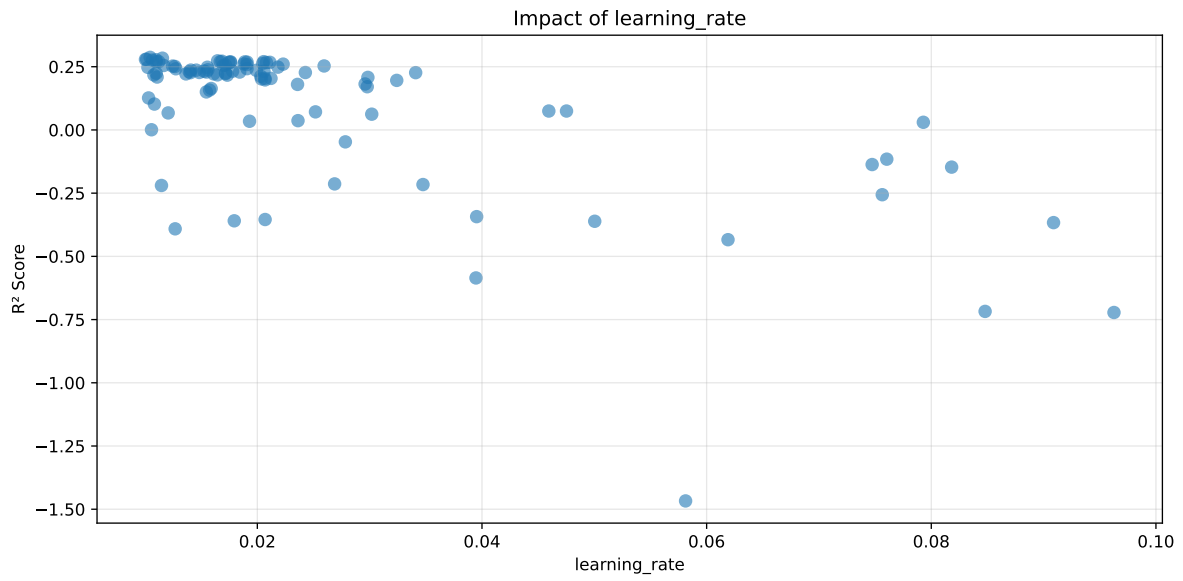
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

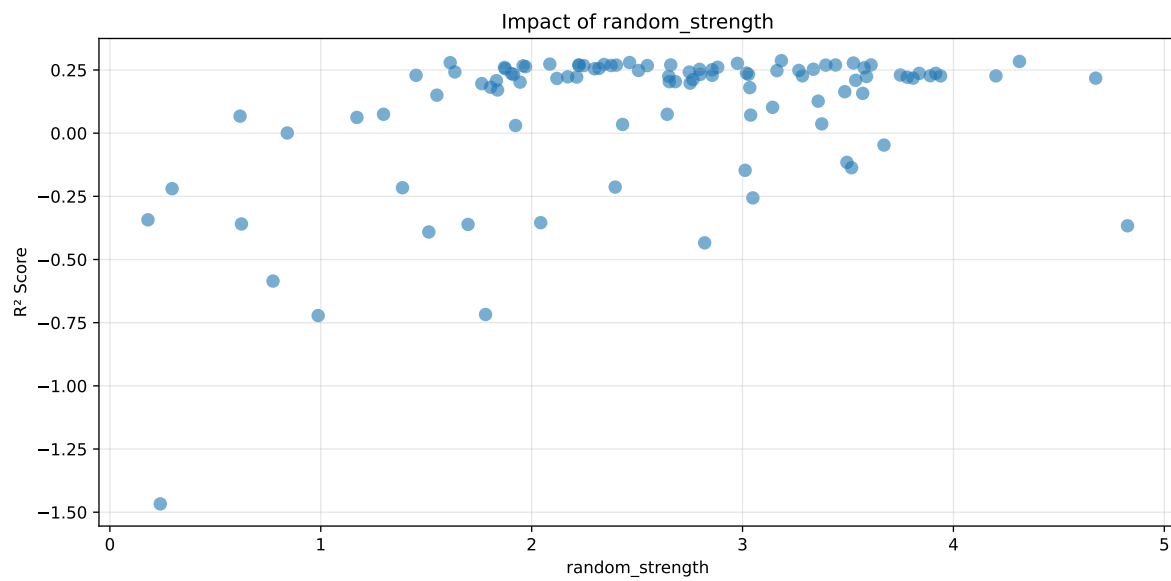
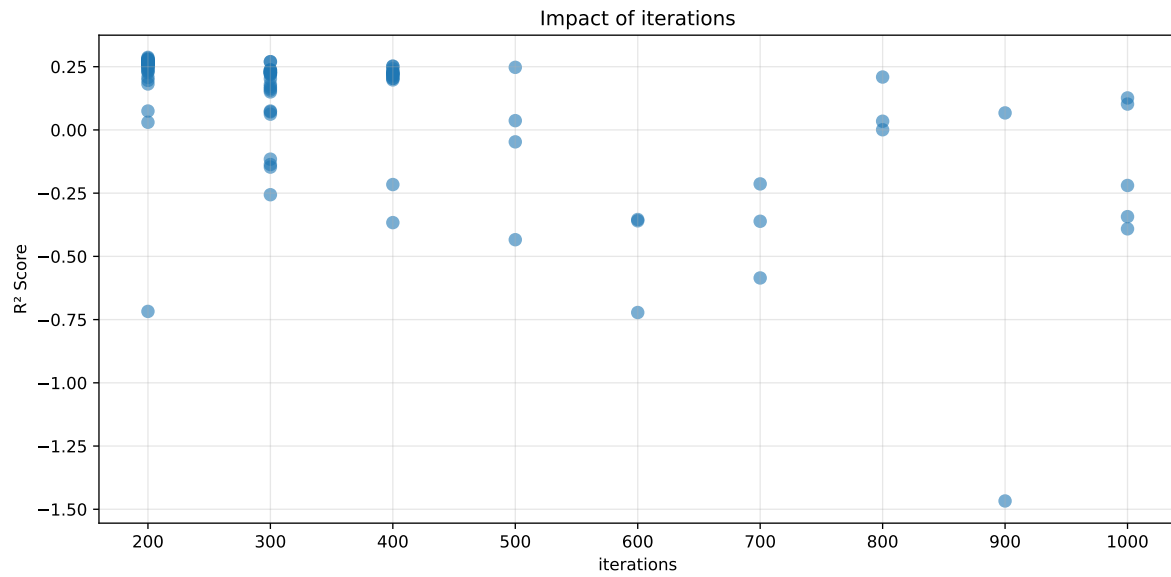
Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813
y_test: mean=2.212, std=0.659, min=1.071, max=4.858

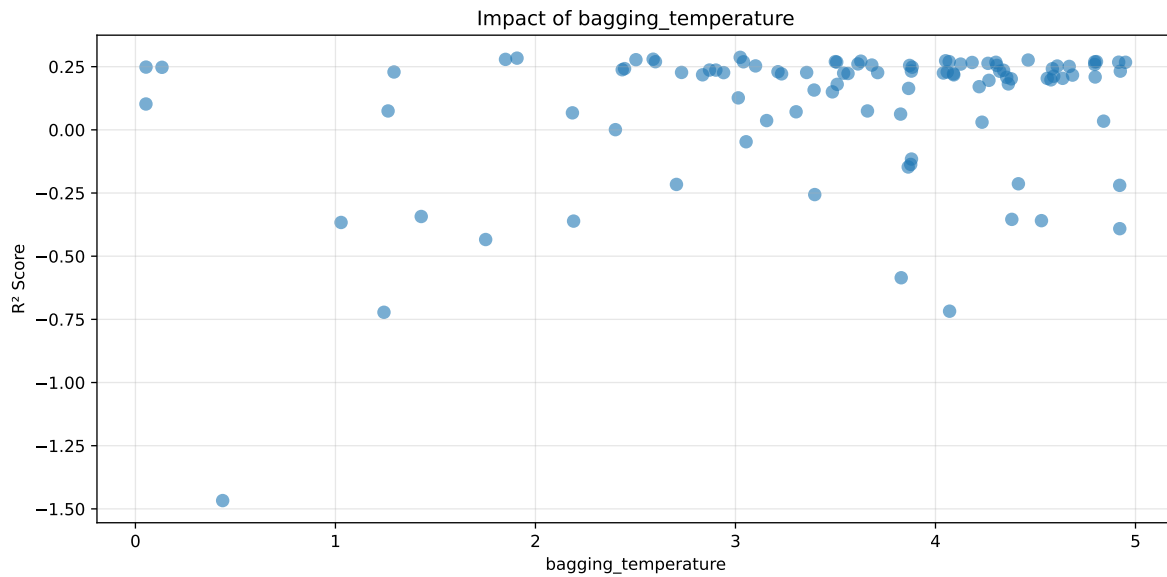
===== TOP Importance FEATURES =====
feature importance r_umidity 65.111582 r_temperature 11.966346 r_NH3 9.229218 r_CO2
8.139509 r_THI 5.553344



Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'iterations', 'random_strength', 'bagging_temperature']

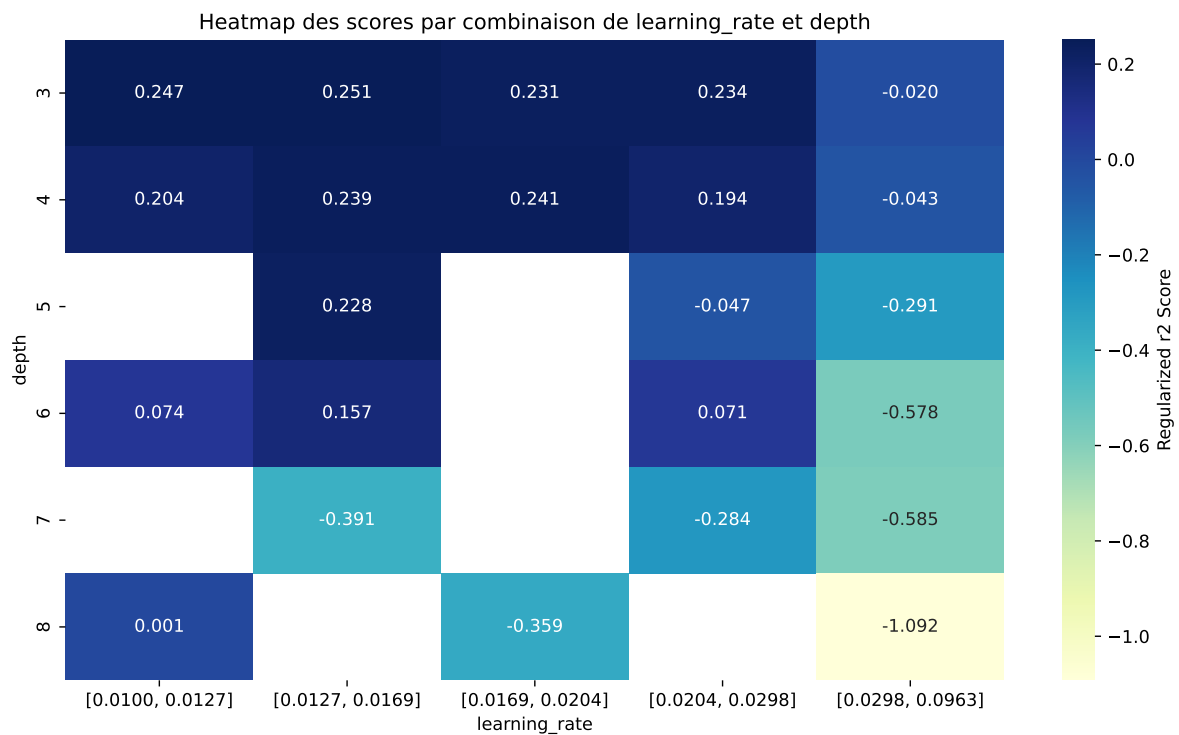






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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.443	0.417	0.474	0.0219	0.287	0.4774	0.0604	1.1449	1.0624
2	0.43	0.4057	0.4658	0.0209	0.2838	0.4664	0.0608	1.1498	1.0601
3	0.4293	0.4043	0.4636	0.0193	0.2796	0.4668	0.0625	1.1546	1.0617
4	0.4359	0.4097	0.4673	0.0207	0.2789	0.4718	0.0621	1.1516	1.0639
5	0.4297	0.4044	0.464	0.0197	0.2776	0.463	0.0586	1.1449	1.0627
6	0.4332	0.4071	0.4661	0.0197	0.2763	0.4631	0.056	1.1377	1.0643
7	0.4609	0.4297	0.4934	0.0231	0.2733	0.4951	0.0655	1.1523	1.0728
8	0.461	0.4295	0.4912	0.0222	0.2718	0.4983	0.0688	1.1603	1.0734
9	0.4667	0.4339	0.4961	0.0219	0.27	0.5	0.0661	1.1524	1.0755
10	0.4725	0.4388	0.5012	0.0235	0.27	0.5053	0.0665	1.1515	1.0769
---	---	---	---	---	---	---	---	---	---
100	0.9655	0.5601	0.6002	0.0357	-1.4672	0.6223	0.0623	1.1112	1.7239

Hyperparameters:

Rank 1: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.010458424698582323, 'reg__l2_leaf_reg': 48.19153158046464, 'reg__bagging_temperature': 3.0236326457501557, 'reg__random_strength': 3.184263061174634, 'reg__subsample': 0.5283723385853872, 'reg__min_data_in_leaf': 49, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.7325904773922777, 'reg__border_count': 94, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 3512.244997642779}

Rank 2: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.01155173481987408, 'reg__l2_leaf_reg': 16.34832395775517, 'reg__bagging_temperature': 1.9078148953458451, 'reg__random_strength': 4.313142223632549, 'reg__subsample': 0.5248323897315647, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.7445647857144698, 'reg__border_count': 70, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 3538.514998582296}

Rank 3: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.010144711053000416, 'reg__l2_leaf_reg': 6.1835098984796195, 'reg__bagging_temperature': 2.588344571885972, 'reg__random_strength': 2.4642950830523103, 'reg__subsample': 0.8142111542171712, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.7577700017219259, 'reg__border_count': 50, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 9370.194833086194}

Rank 4: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.01004067555849581, 'reg__l2_leaf_reg': 5.7501654879095, 'reg__bagging_temperature': 1.850248311406053, 'reg__random_strength': 1.6137307208777072, 'reg__subsample': 0.5361091951510167, 'reg__min_data_in_leaf': 47, 'reg__leaf_estimation_iterations': 4, 'reg__rsm':

0.7591714998838996, 'reg__border_count': 82, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 3561.098822717765}

Rank 5: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.010982145682204215, 'reg__l2_leaf_reg': 5.843970145134149, 'reg__bagging_temperature': 2.5023923471346468, 'reg__random_strength': 3.525594455810416, 'reg__subsample': 0.5325491533759992, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.7481674635577029, 'reg__border_count': 71, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 3439.635739117579}

Rank 6: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.010658415478409832, 'reg__l2_leaf_reg': 2.1753788488403707, 'reg__bagging_temperature': 4.463298774391951, 'reg__random_strength': 2.975813550887133, 'reg__subsample': 0.6195758114060552, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.6621634114700313, 'reg__border_count': 58, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 9739.010977270427}

Rank 7: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.016464115735609502, 'reg__l2_leaf_reg': 31.437237306473307, 'reg__bagging_temperature': 4.050703340277355, 'reg__random_strength': 2.0861615508256066, 'reg__subsample': 0.5936113204690991, 'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.8012208156317933, 'reg__border_count': 117, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 6282.451335864584}

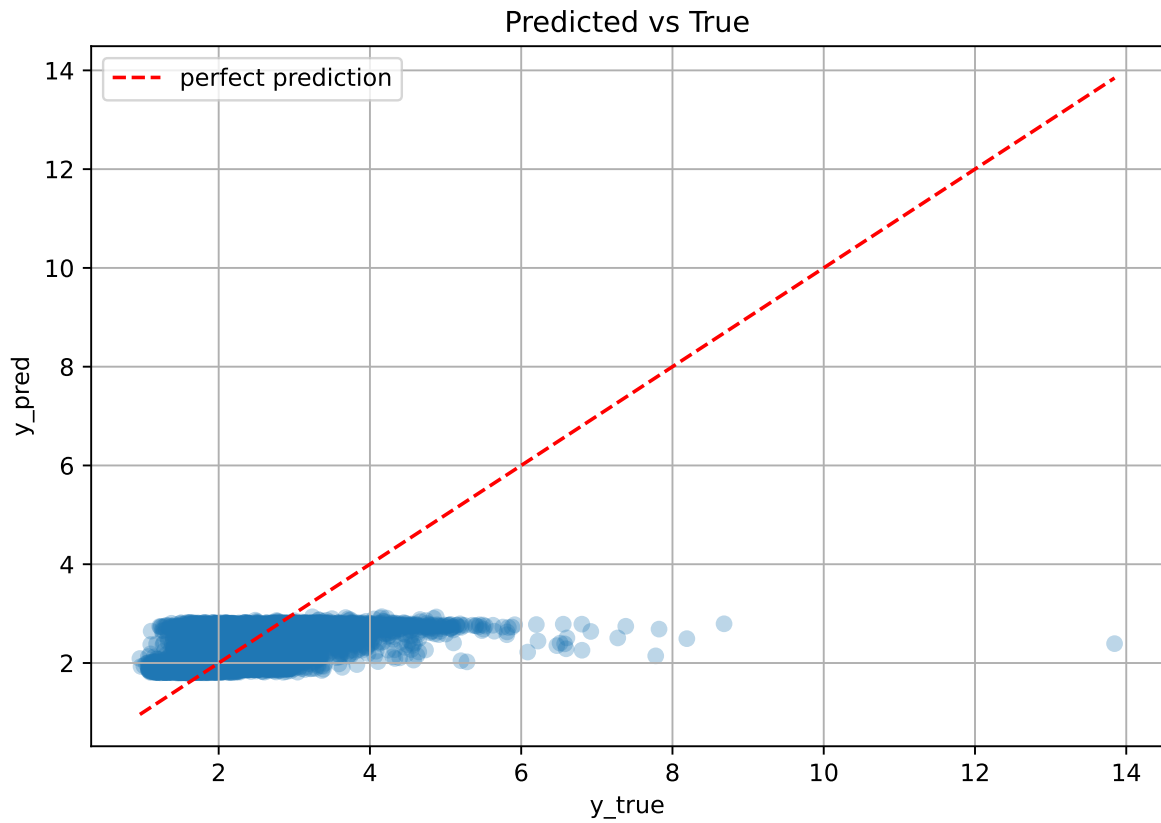
Rank 8: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.016853799598180782, 'reg__l2_leaf_reg': 33.81733596912122, 'reg__bagging_temperature': 3.6267697990210777, 'reg__random_strength': 2.3431047332900206, 'reg__subsample': 0.5972438490533294, 'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.800857787284592, 'reg__border_count': 111, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4353.943874697178}

Rank 9: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.016696421078014733, 'reg__l2_leaf_reg': 34.788967221603535, 'reg__bagging_temperature': 4.804845061528546, 'reg__random_strength': 2.2230879912450914, 'reg__subsample': 0.5861239585606297, 'reg__min_data_in_leaf': 34, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.7901402649430236, 'reg__border_count': 111, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4452.476188323182}

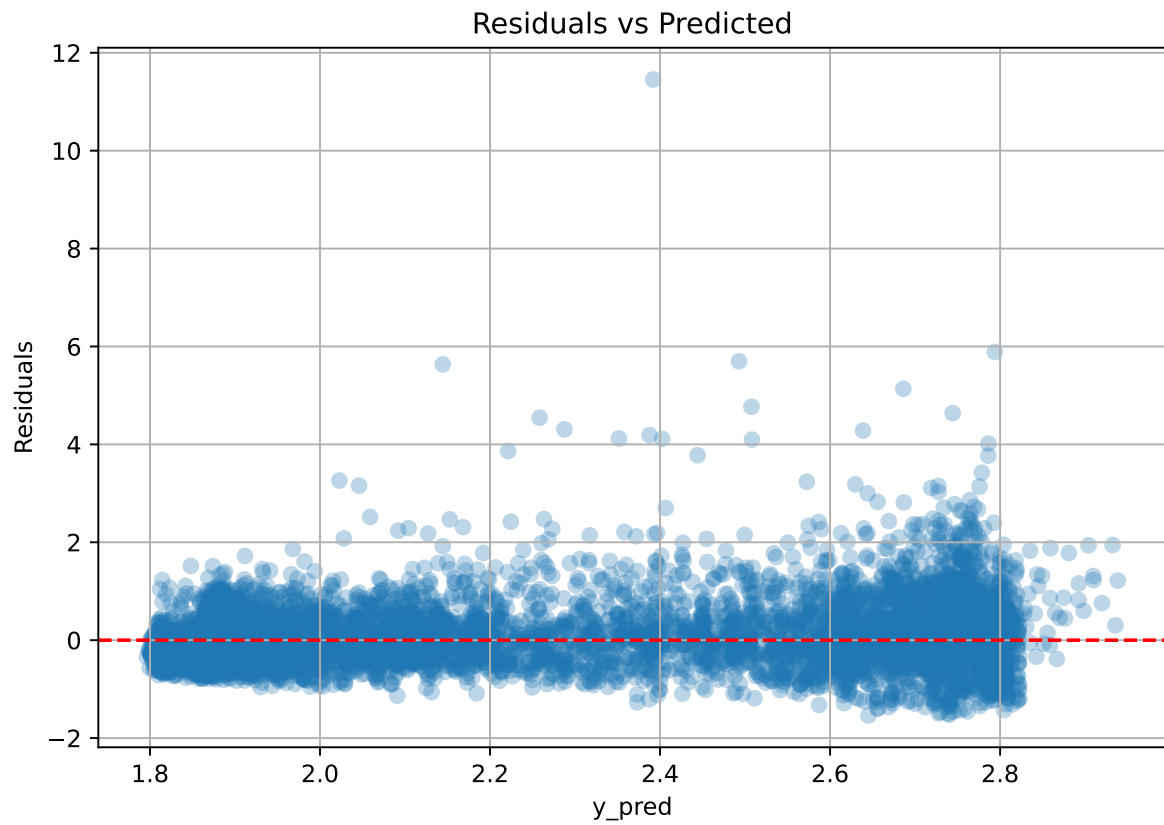
Rank 10: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.020564378072751716, 'reg__l2_leaf_reg': 30.848566430324684, 'reg__bagging_temperature': 4.068609945078723,


```
'reg__random_strength': 2.6595940719000644, 'reg__subsample': 0.6074208140341114,  
'reg__min_data_in_leaf': 28, 'reg__leaf_estimation_iterations': 2, 'reg__rsm':  
0.8213117937510848, 'reg__border_count': 121, 'reg__leaf_estimation_method': 'Newton',  
'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_tempera-  
ture': 3977.378466824856}
```

2.3.10 Scatter



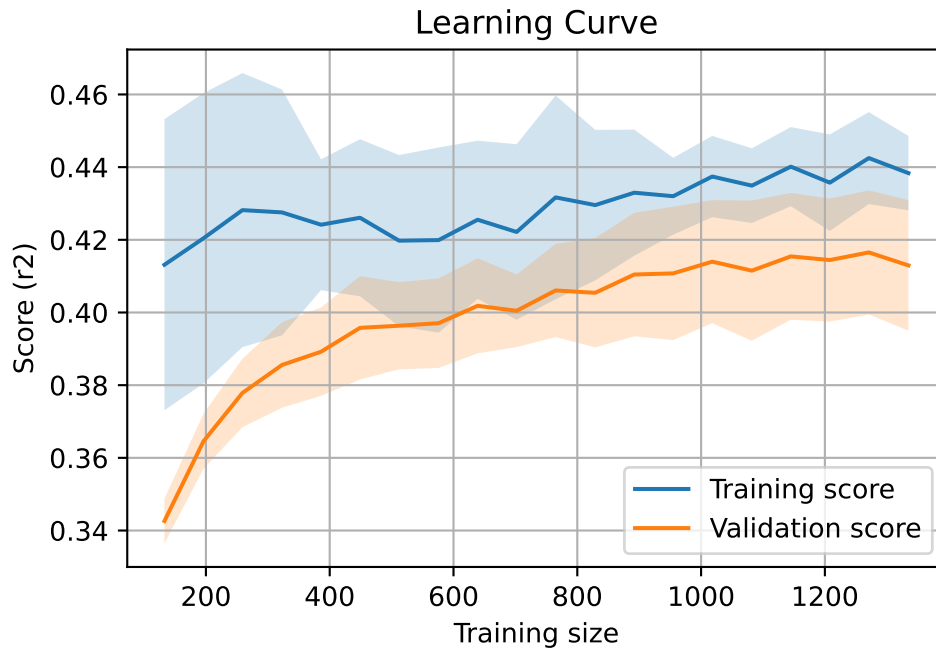
2.3.11 Scatter residuals



2.3.12 Learning curve — Group: 5

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817

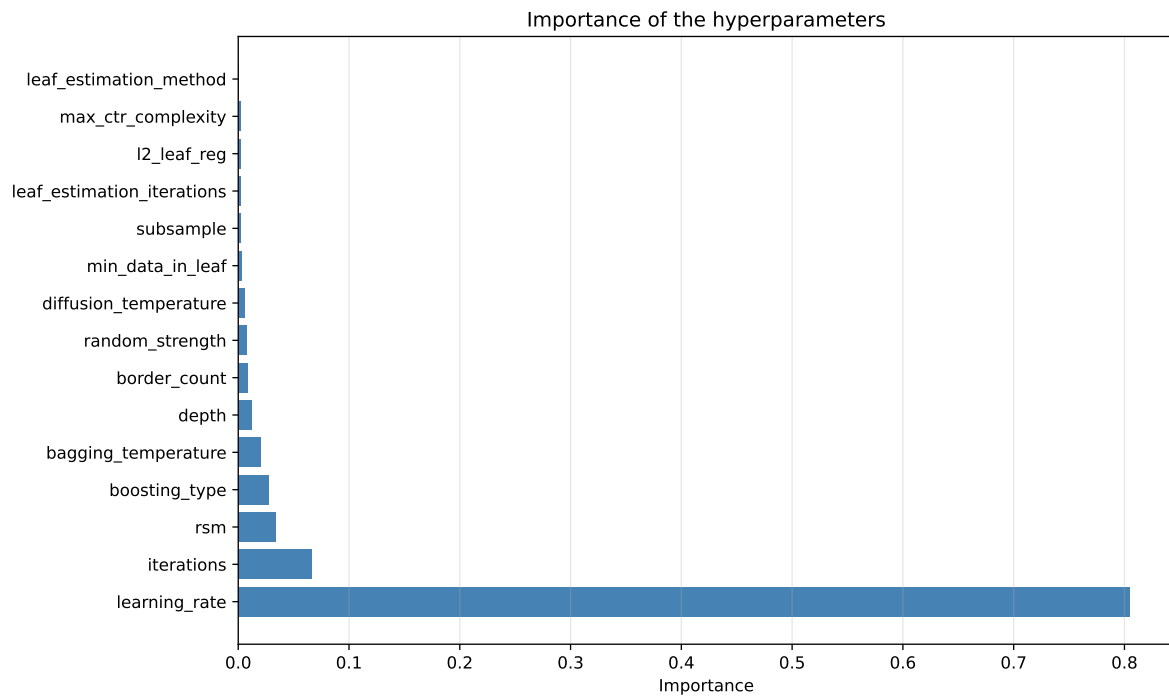
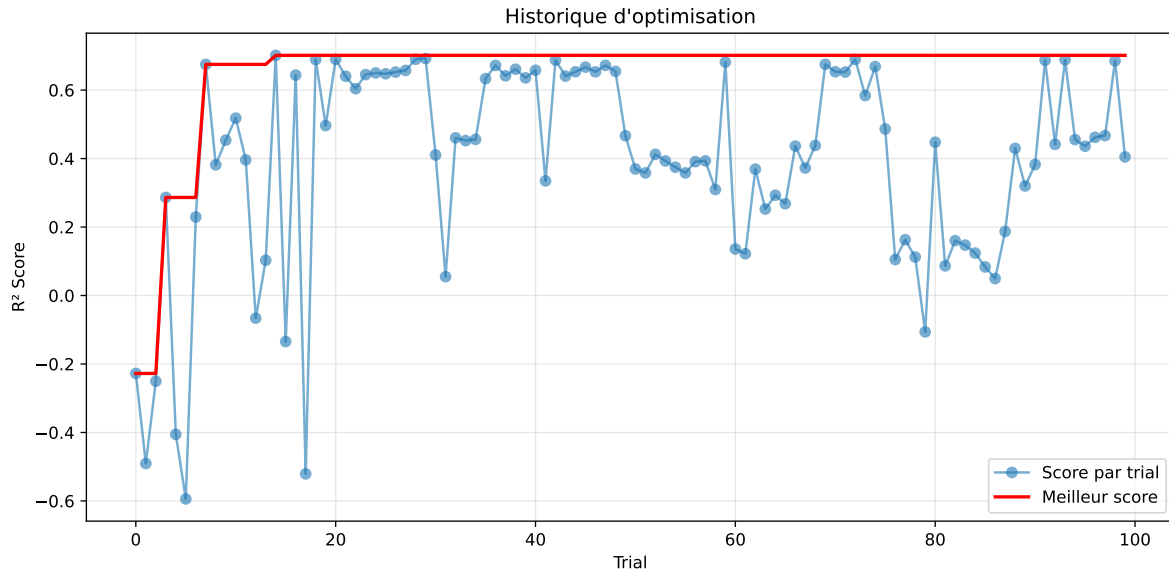


2.3.13 Gridsearch — Group: 11

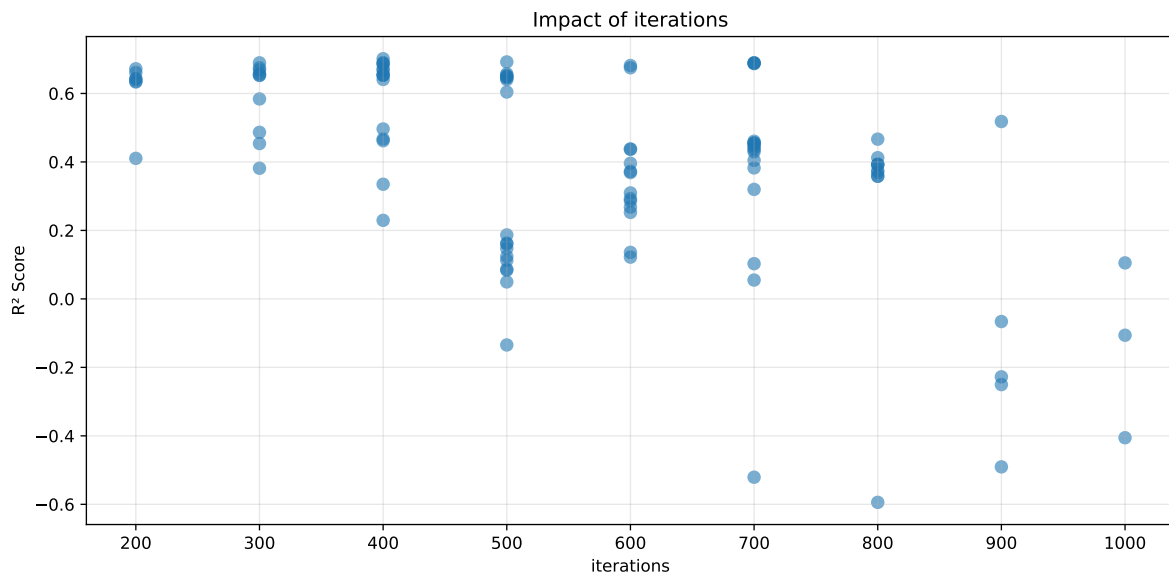
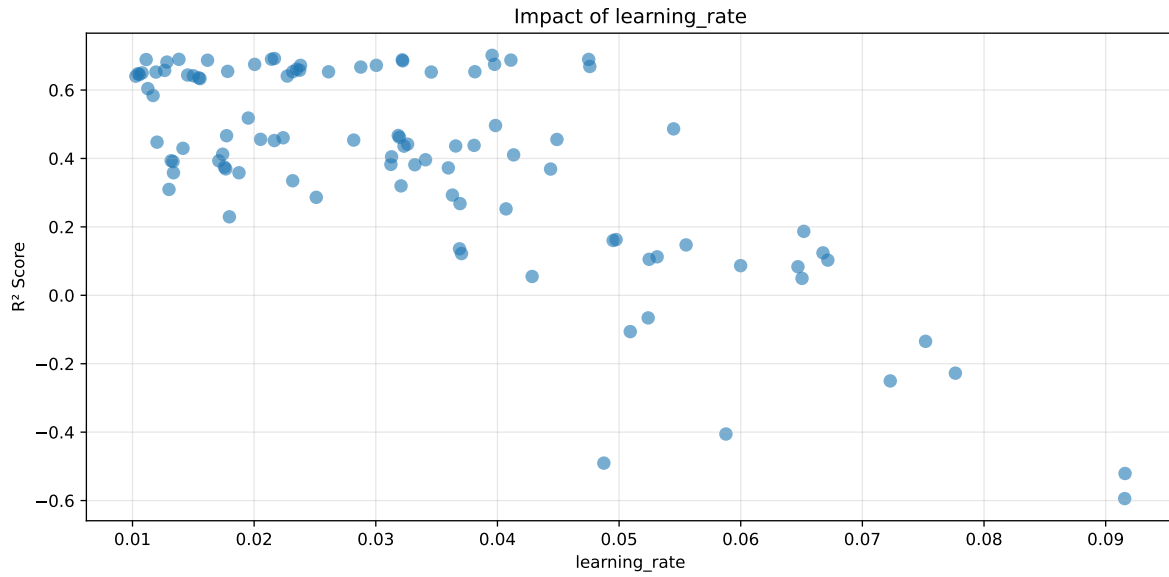
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682
y_test: mean=2.561, std=0.947, min=1.120, max=13.845

===== TOP Importance FEATURES =====
feature importance r_umidity 32.521813 r_NH3 28.452990 r_temperature 15.825210 r_CO2
14.028317 r_THI 9.171670

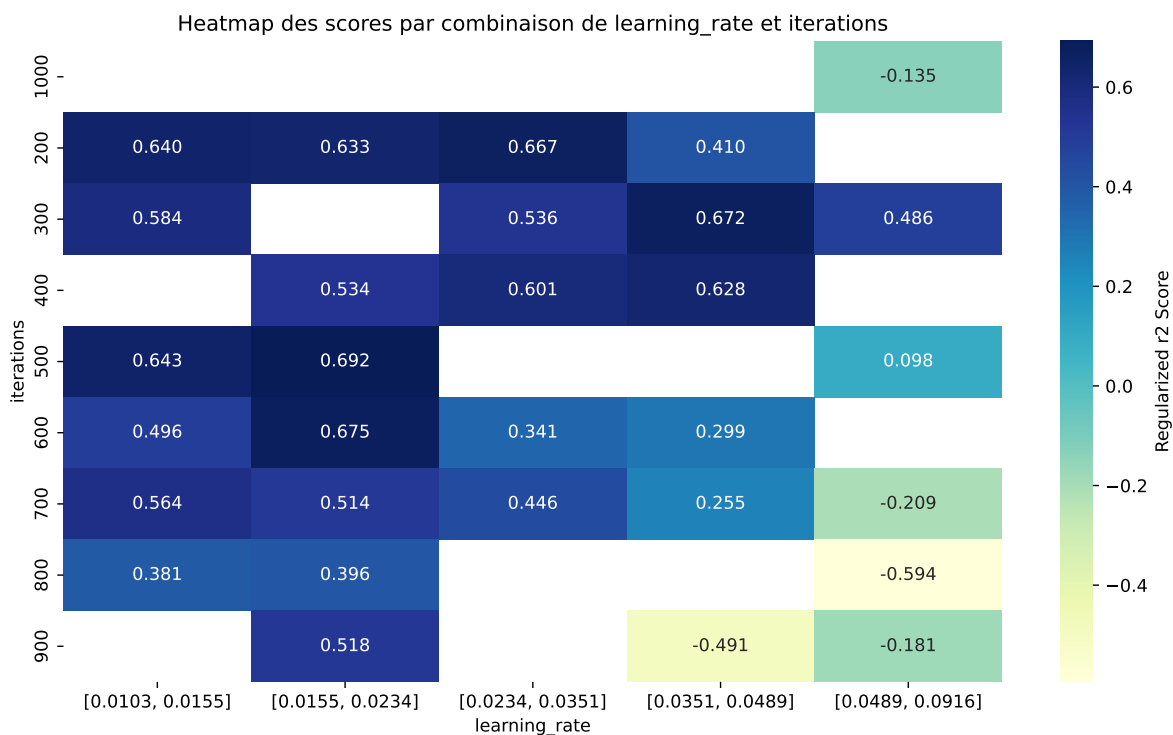


Paramètres avec >5% d'importance : ['learning_rate', 'iterations']



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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7325	0.7014	0.5922	0.0264	0.7014	0.5781	-0.1234	0.8241	1.0442
2	0.7194	0.6921	0.5982	0.0207	0.6921	0.6008	-0.0913	0.868	1.0394
3	0.7191	0.6901	0.594	0.0189	0.6901	0.5955	-0.0946	0.8629	1.0421
4	0.7178	0.6898	0.5959	0.0159	0.6898	0.6017	-0.0881	0.8722	1.0406
5	0.7196	0.6895	0.5862	0.0148	0.6895	0.5964	-0.0931	0.865	1.0437
6	0.7178	0.6891	0.5939	0.0164	0.6891	0.6	-0.0891	0.8707	1.0416
7	0.7205	0.6883	0.5939	0.017	0.6883	0.5981	-0.0903	0.8689	1.0468
8	0.7108	0.6875	0.591	0.0222	0.6875	0.5956	-0.0918	0.8664	1.0339
9	0.7104	0.6871	0.5901	0.0165	0.6871	0.5991	-0.088	0.8719	1.0338
10	0.7094	0.6853	0.5908	0.0186	0.6853	0.5981	-0.0872	0.8728	1.0351
---	---	---	---	---	---	---	---	---	---
100	0.9573	0.6987	0.5829	0.0238	-0.5942	0.5812	-0.1175	0.8319	1.3701

Hyperparameters:

Rank 1: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.03956245923953242, 'reg__l2_leaf_reg': 4.796303732313237, 'reg__bagging_temperature': 0.639652888635549, 'reg__random_strength': 4.361235747404775, 'reg__subsample': 0.8029122857171798, 'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6413601417856691, 'reg__border_count': 100, 'reg__leaf_estimation_method': 'Gradient',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 7701.819412987404}

Rank 2: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.02166763539883479, 'reg__l2_leaf_reg': 15.791602123184383, 'reg__bagging_temperature': 0.947890321121126, 'reg__random_strength': 4.23948771122261, 'reg__subsample': 0.785280823858325, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.7758375198073945, 'reg__border_count': 101, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 8280.033916922048}

Rank 3: {'reg__iterations': 400, 'reg__depth': 4, 'reg__learning_rate': 0.02142603466527776, 'reg__l2_leaf_reg': 1.4400251751628654, 'reg__bagging_temperature': 0.8368437712577312, 'reg__random_strength': 3.9006037303494745, 'reg__subsample': 0.8066448319675885, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.7558147222309992, 'reg__border_count': 104, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 7953.2393148875835}

Rank 4: {'reg__iterations': 700, 'reg__depth': 5, 'reg__learning_rate': 0.013815756775309909, 'reg__l2_leaf_reg': 41.751391015815834, 'reg__bagging_temperature': 2.33226644022071, 'reg__random_strength': 3.587950269538045, 'reg__subsample': 0.7335825417498048, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.5881006393480256, 'reg__border_count': 122, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5934.275816255145}

Rank 5: {'reg__iterations': 300, 'reg__depth': 6, 'reg__learning_rate': 0.04749855522672043, 'reg__l2_leaf_reg': 33.090615156635835, 'reg__bagging_temperature': 2.3960961712011946, 'reg__random_strength': 4.407519163849412, 'reg__subsample': 0.6100298419909961, 'reg__min_data_in_leaf': 11, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.5118689094991666, 'reg__border_count': 36, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 4191.9992531909165}

Rank 6: {'reg__iterations': 700, 'reg__depth': 5, 'reg__learning_rate': 0.011123126418891162, 'reg__l2_leaf_reg': 35.0978752540466, 'reg__bagging_temperature': 2.6117386253398003, 'reg__random_strength': 1.1008930508286614, 'reg__subsample': 0.6687325345976556, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.5682233882979946, 'reg__border_count': 36, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 1772.491028130645}

Rank 7: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.032177440779485235, 'reg__l2_leaf_reg': 35.41891103757431, 'reg__bagging_temperature': 1.7732244496258787, 'reg__random_strength': 3.973116721755786, 'reg__subsample': 0.7722742319587448,

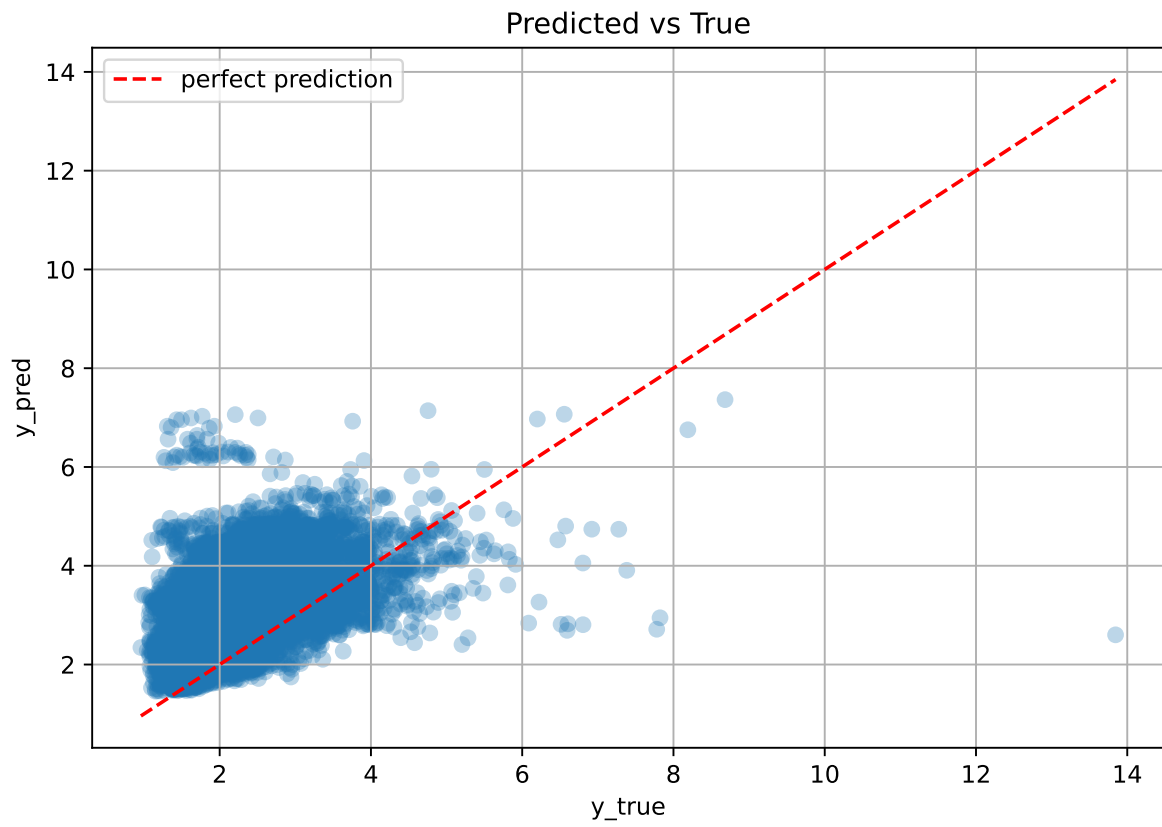
'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.7636305232032188, 'reg__border_count': 44, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 9300.503897408711}

Rank 8: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.04110657394328352, 'reg__l2_leaf_reg': 27.572585442282822, 'reg__bagging_temperature': 0.6955606686003747, 'reg__random_strength': 4.532869086332714, 'reg__subsample': 0.8044711146868957, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6782445836648184, 'reg__border_count': 127, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 9951.554855821414}

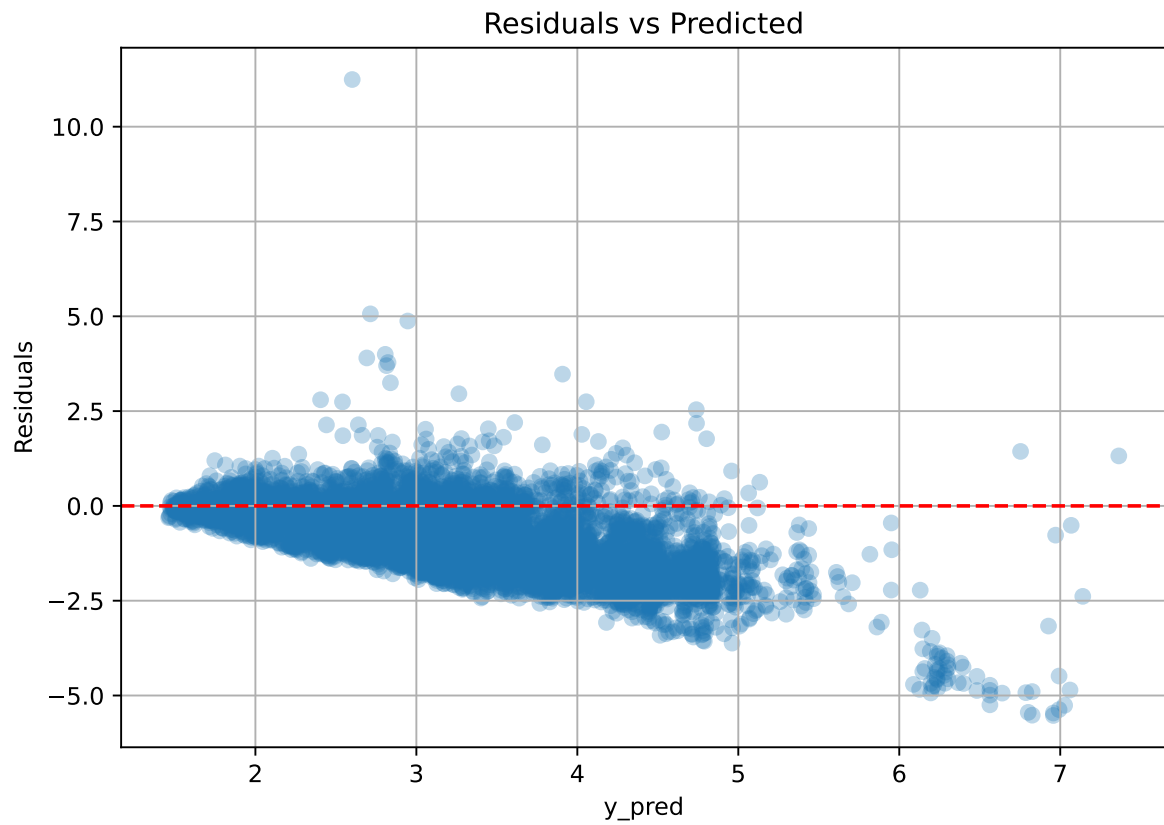
Rank 9: {'reg__iterations': 700, 'reg__depth': 5, 'reg__learning_rate': 0.016168743872492702, 'reg__l2_leaf_reg': 37.65085260818245, 'reg__bagging_temperature': 1.763345499417049, 'reg__random_strength': 2.9826526226412877, 'reg__subsample': 0.8271434087465606, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.5652140552158903, 'reg__border_count': 122, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 276.86151374481733}

Rank 10: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.03221280503139448, 'reg__l2_leaf_reg': 47.7666894271625, 'reg__bagging_temperature': 1.8544811502195104, 'reg__random_strength': 2.3279486800371023, 'reg__subsample': 0.5638769511709375, 'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.7588574210893418, 'reg__border_count': 42, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 9081.86812393614}

2.3.14 Scatter



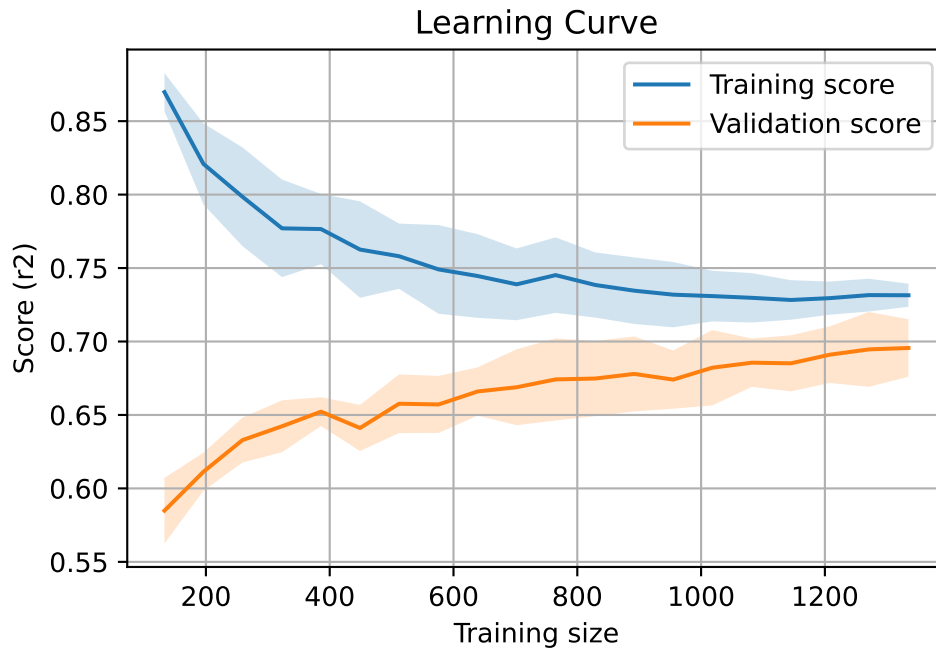
2.3.15 Scatter residuals



2.3.16 Learning curve — Group: 11

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817



2.4 Sumup-table

model	group	Len training	Len test	Len total	mean_train_cv	mean_test_cv	test_cv	std_test_cv	pen_score	test	diff	r_test	r_train
CatBoost	3	950	406	1356	0.622	0.566	0.637	0.039	0.2857	0.646	0.08	1.142	1.099
CatBoost	4	952	408	1360	0.342	0.31	0.332	0.039	0.149	0.341	0.031	1.101	1.104
CatBoost	5	1909	817	2726	0.443	0.417	0.474	0.022	0.287	0.477	0.06	1.145	1.062
CatBoost	11	1909	817	2726	0.732	0.701	0.592	0.026	0.7014	0.578	-0.123	0.824	1.044
CatBoost	MEAN	5720	2448	8168	0.535	0.499	0.509	0.032	0.356	0.511	0.012	1.053	1.077

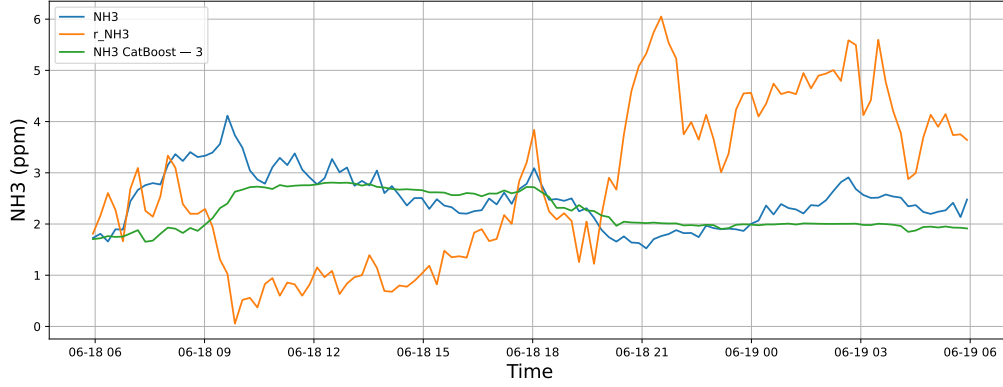
3 Plot

3.1 Standalone plots

3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

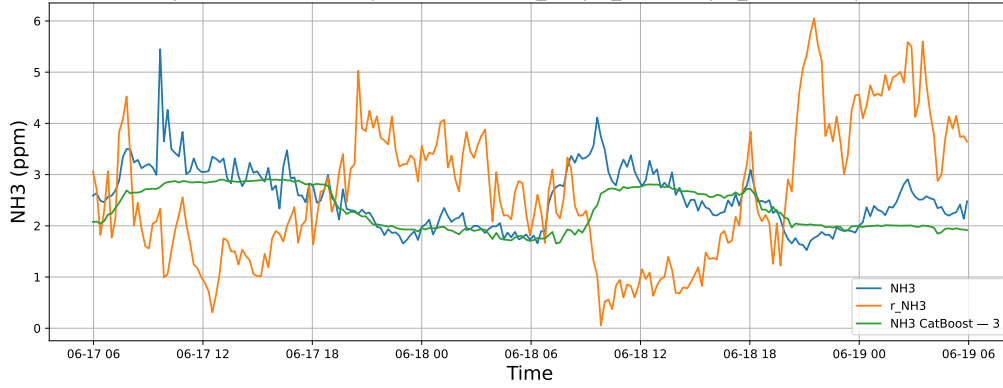
3.1.1.1 Time window : 24

NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



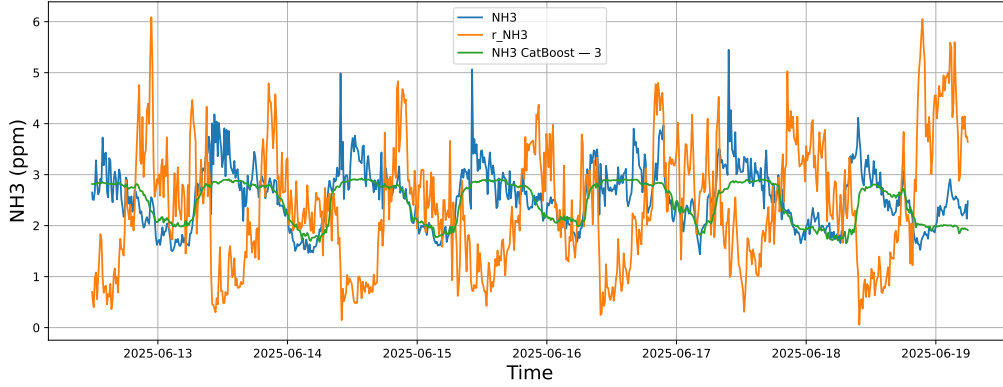
3.1.1.2 Time window : 48

NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.1.1.3 Time window : ALL

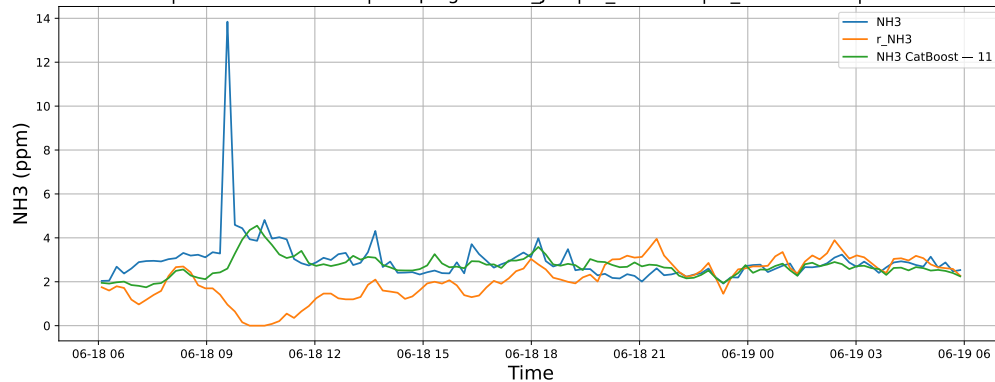
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

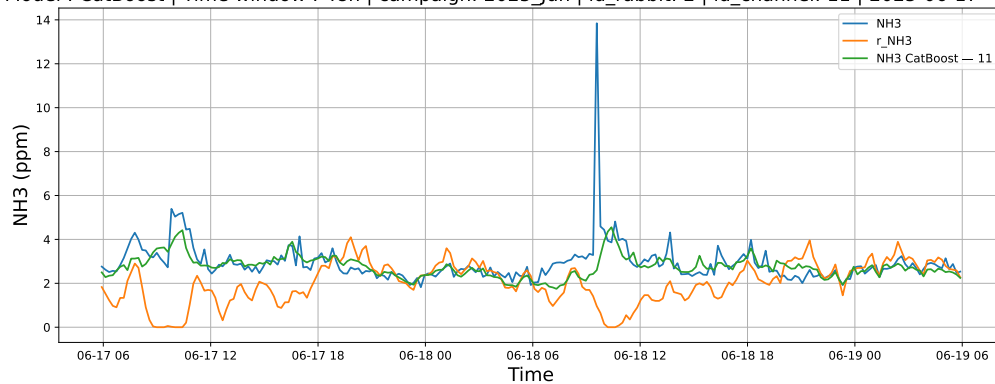
3.1.2.1 Time window : 24

NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



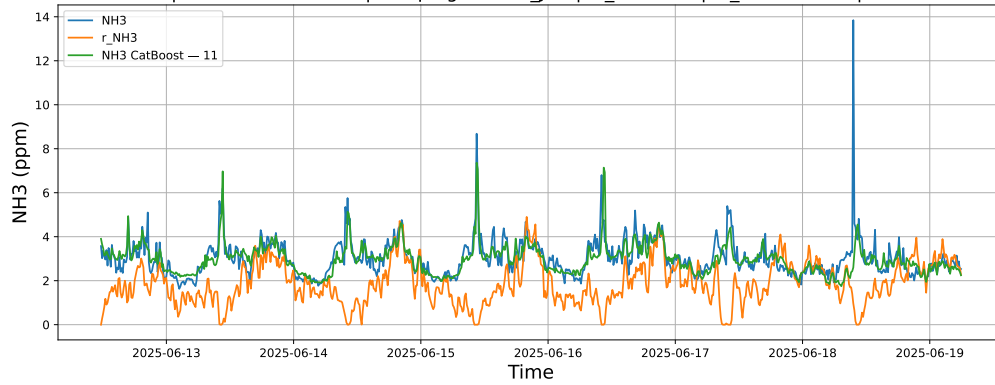
3.1.2.2 Time window : 48

NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.1.2.3 Time window : ALL

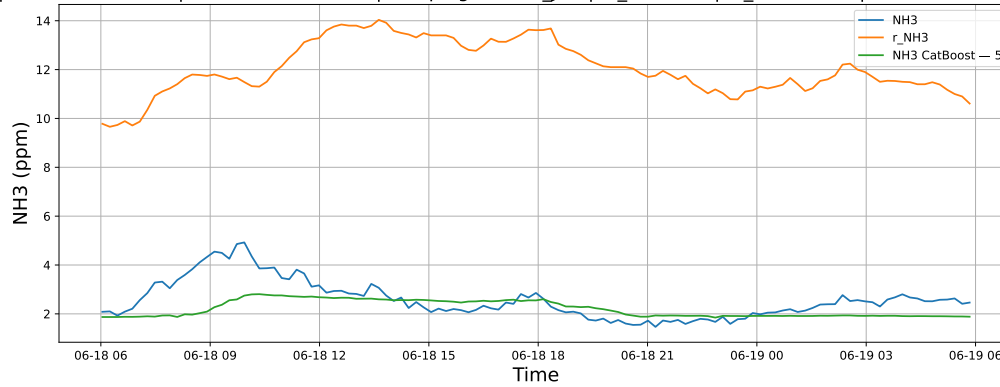
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

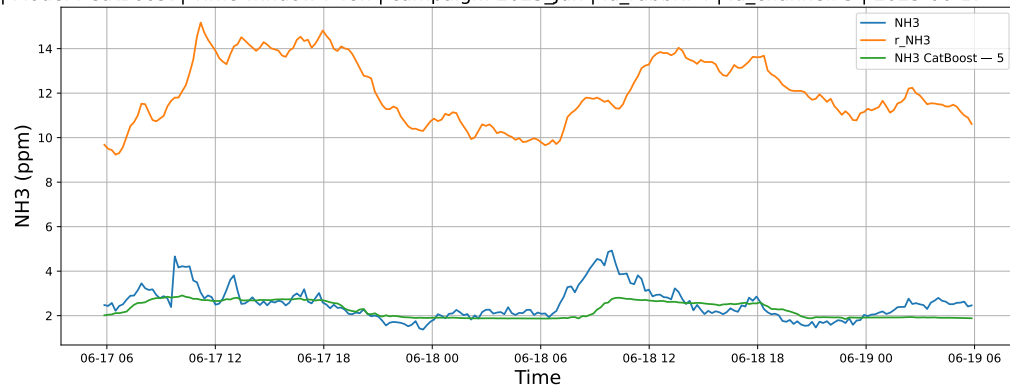
3.1.3.1 Time window : 24

NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



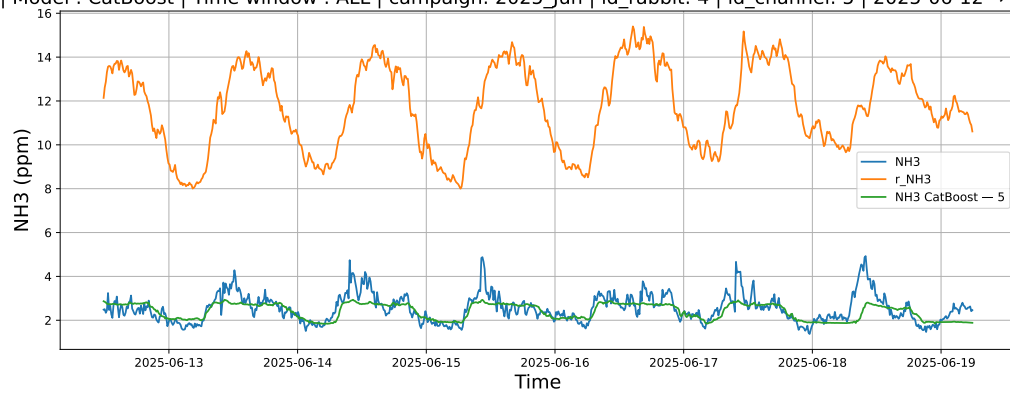
3.1.3.2 Time window : 48

NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.1.3.3 Time window : ALL

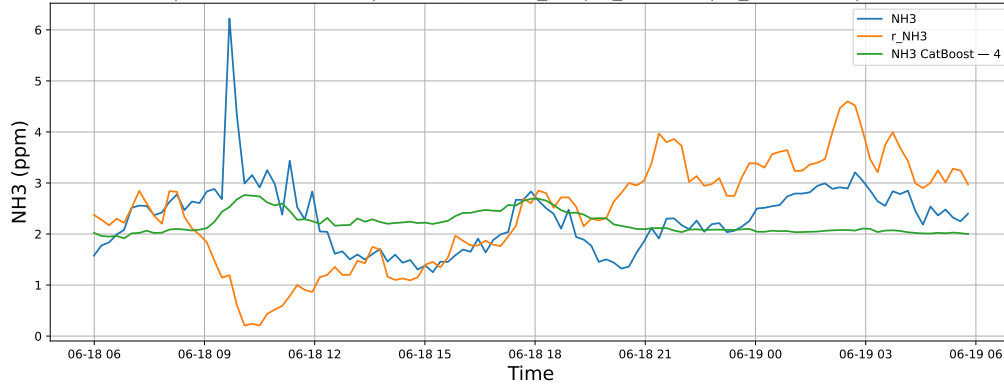
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

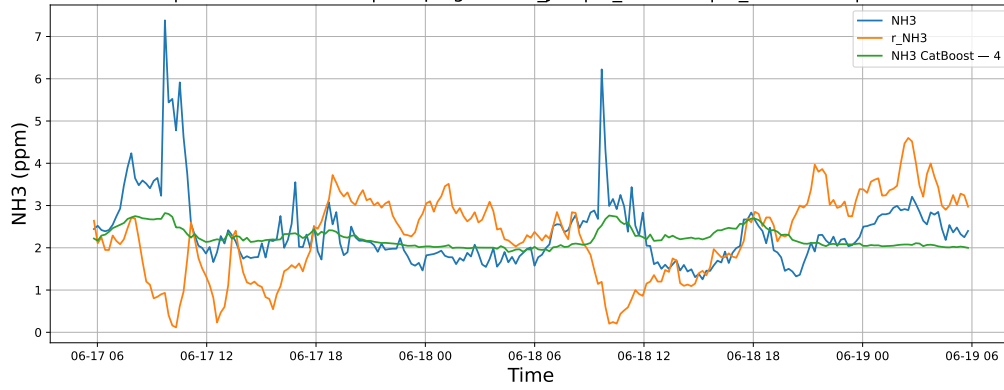
3.1.4.1 Time window : 24

NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



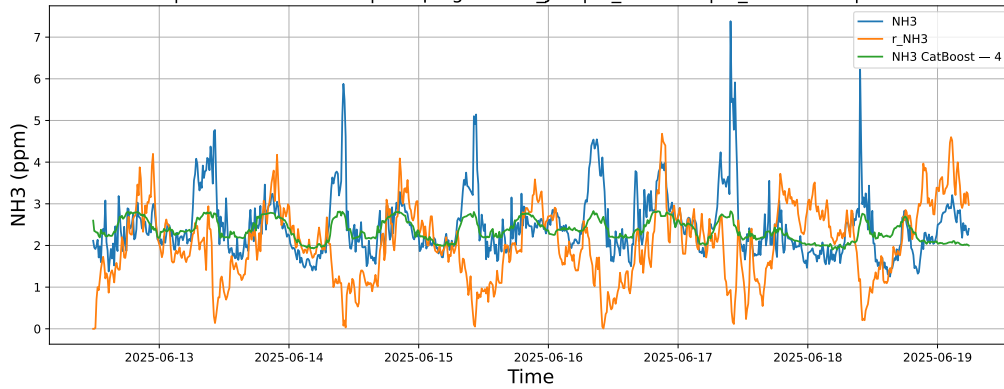
3.1.4.2 Time window : 48

NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.1.4.3 Time window : ALL

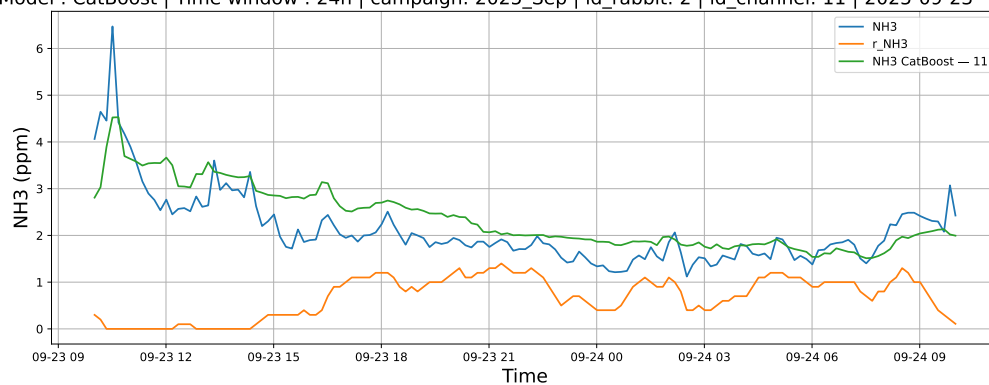
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

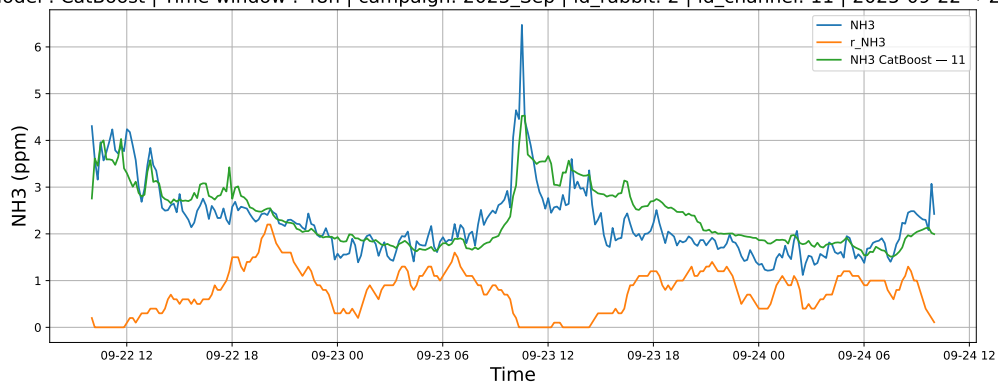
3.1.5.1 Time window : 24

NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



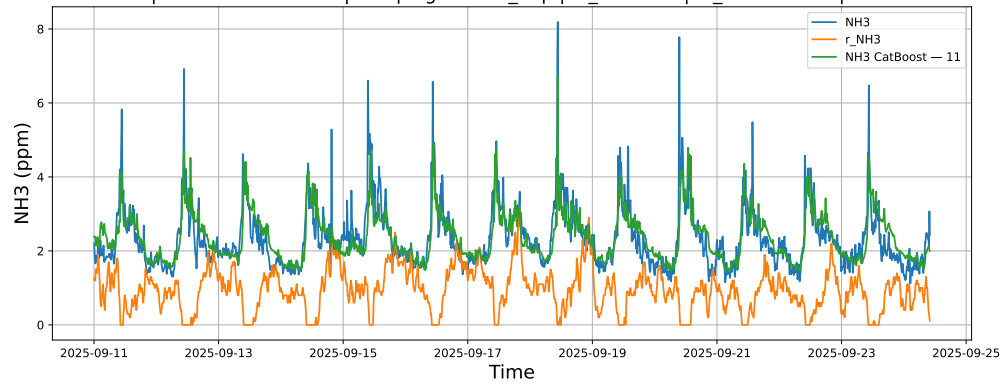
3.1.5.2 Time window : 48

NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.1.5.3 Time window : ALL

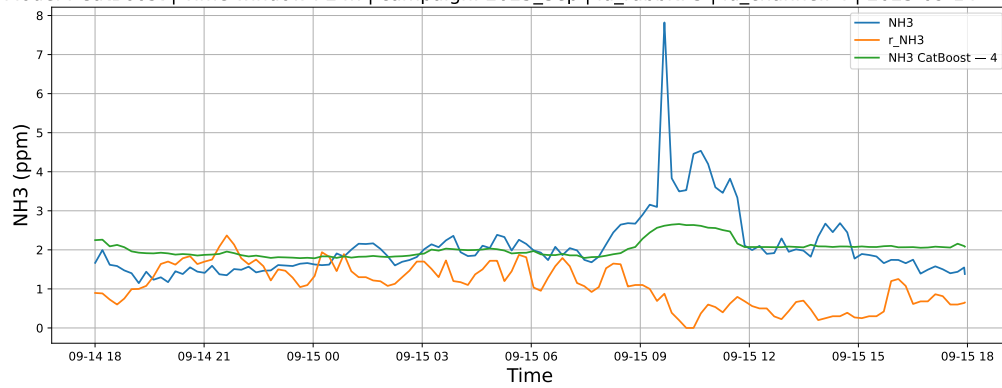
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

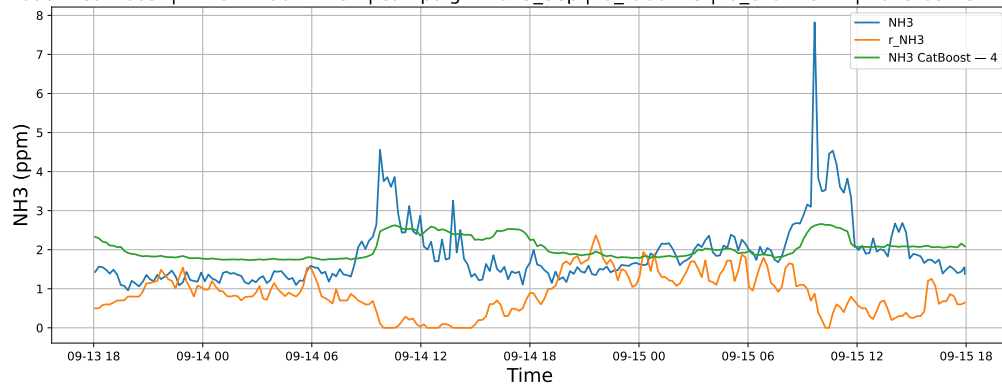
3.1.6.1 Time window : 24

NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



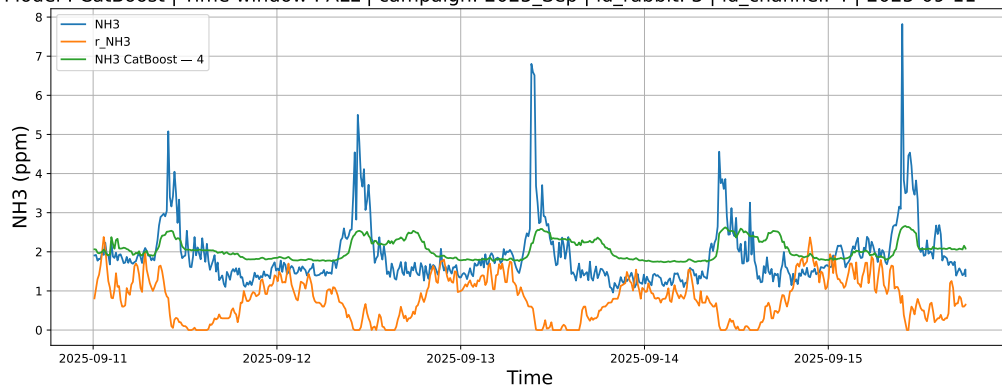
3.1.6.2 Time window : 48

NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.1.6.3 Time window : ALL

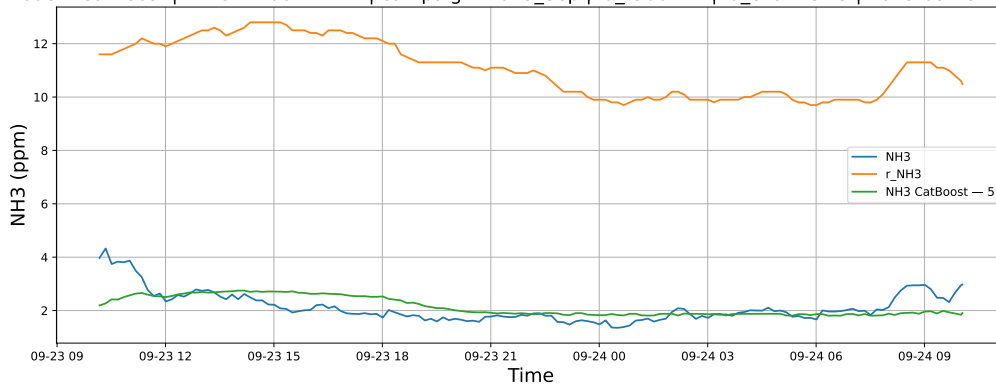
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

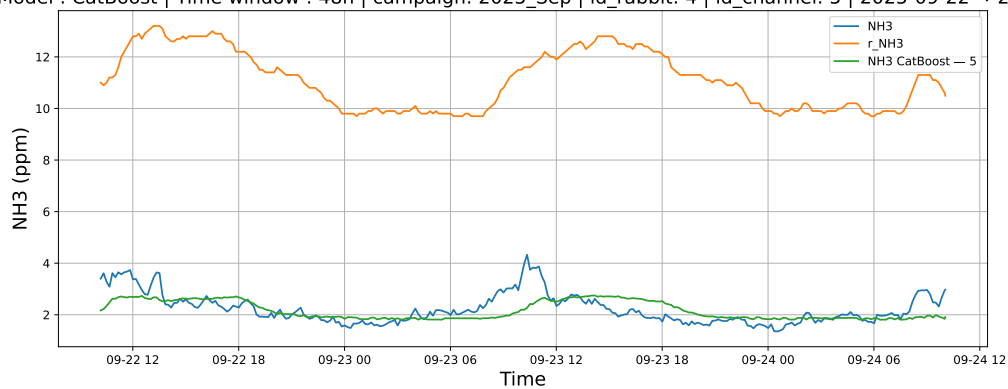
3.1.7.1 Time window : 24

NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



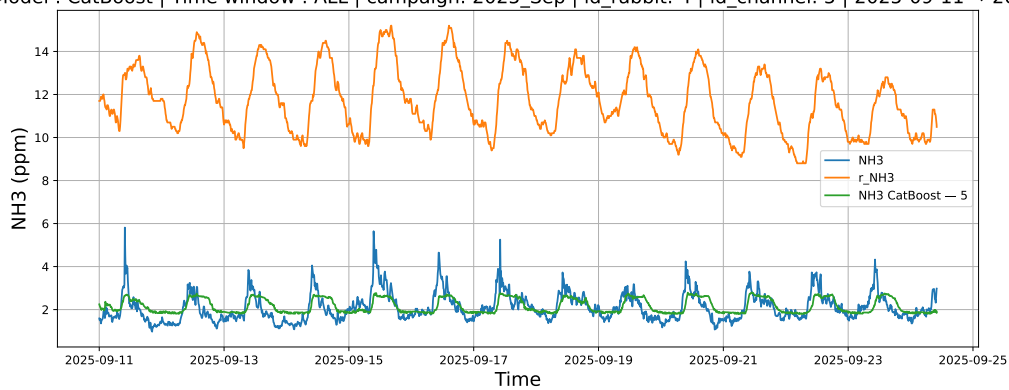
3.1.7.2 Time window : 48

NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.1.7.3 Time window : ALL

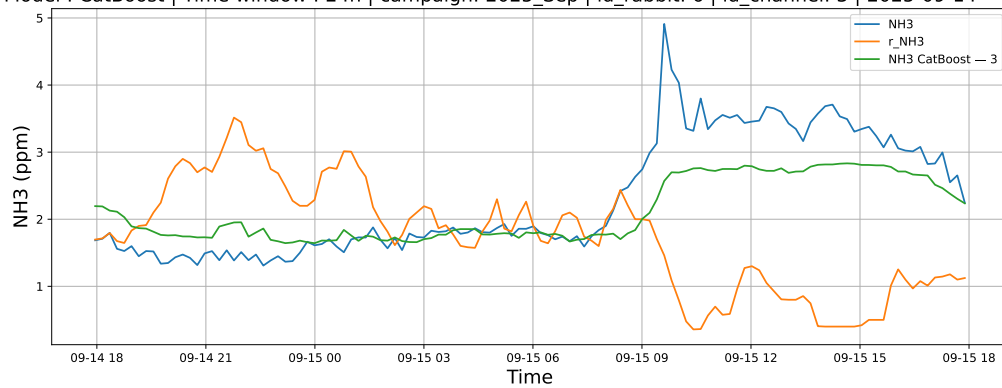
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

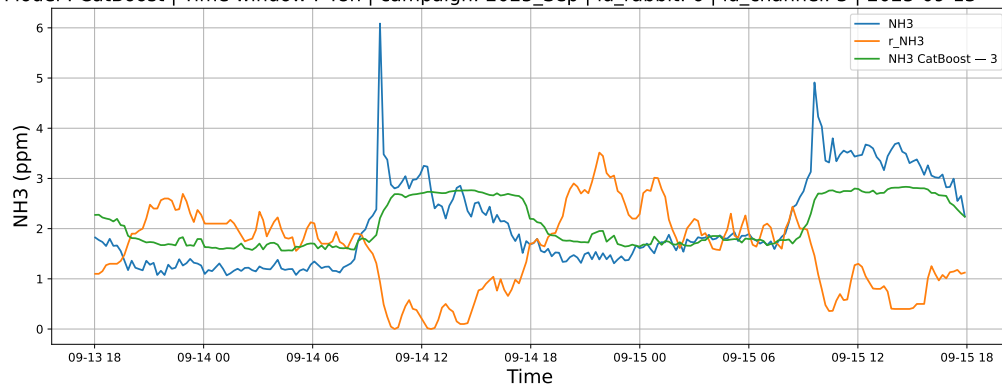
3.1.8.1 Time window : 24

NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



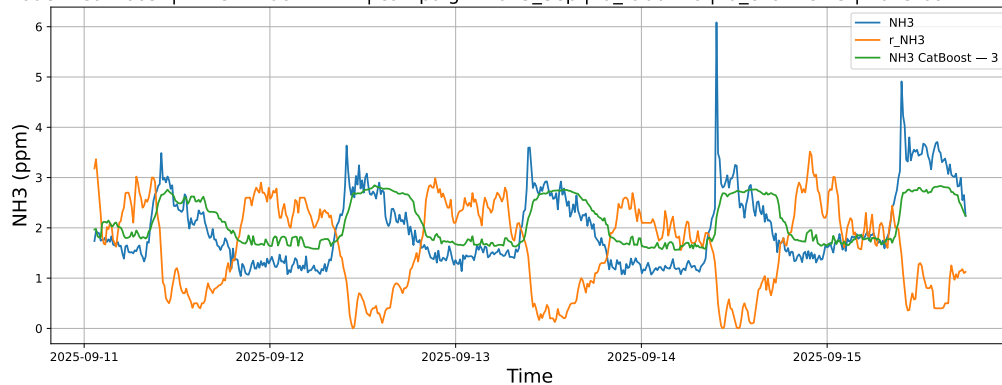
3.1.8.2 Time window : 48

NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.1.8.3 Time window : ALL

NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

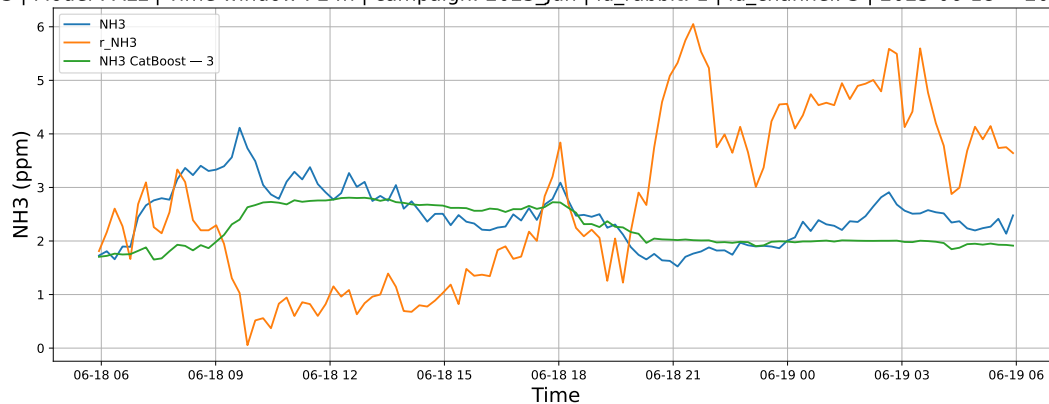


3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

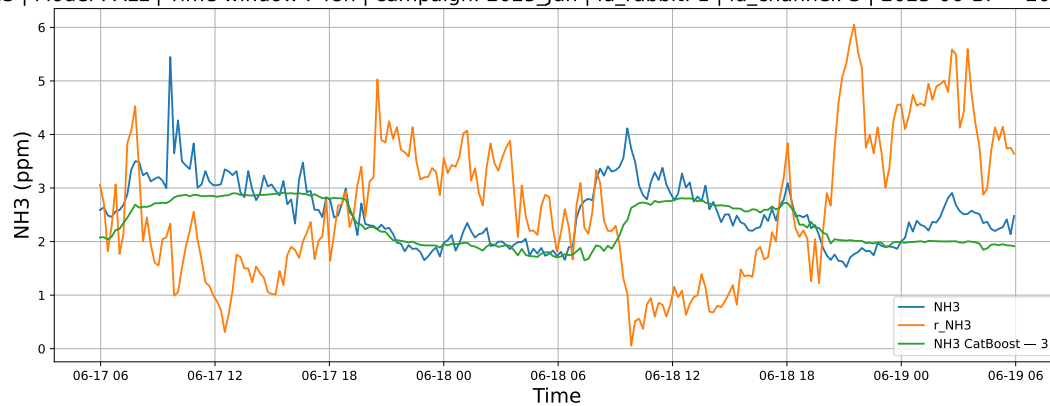
3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



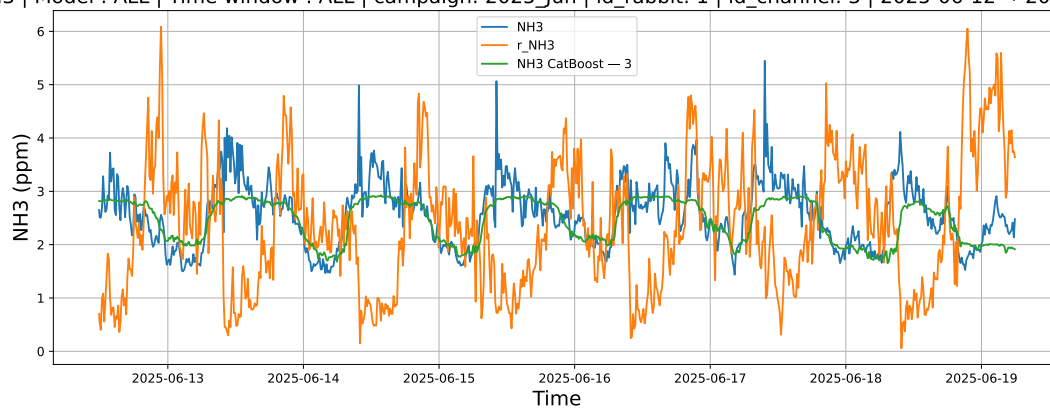
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.2.1.3 Time window : ALL

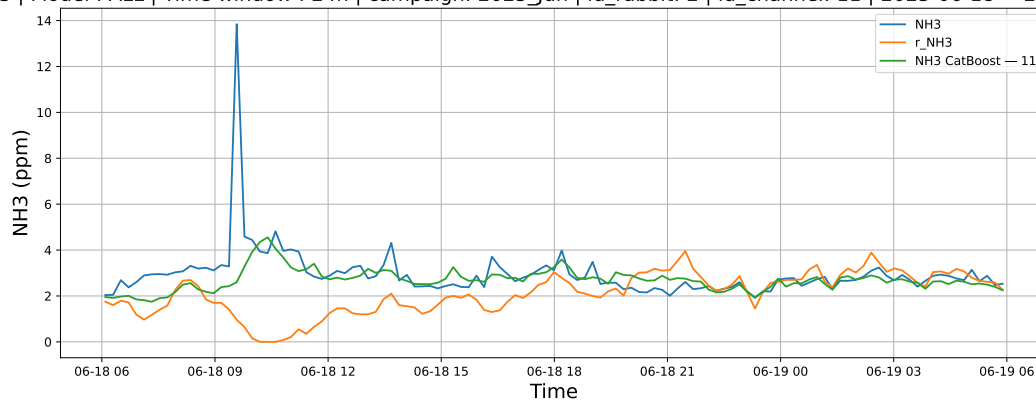
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

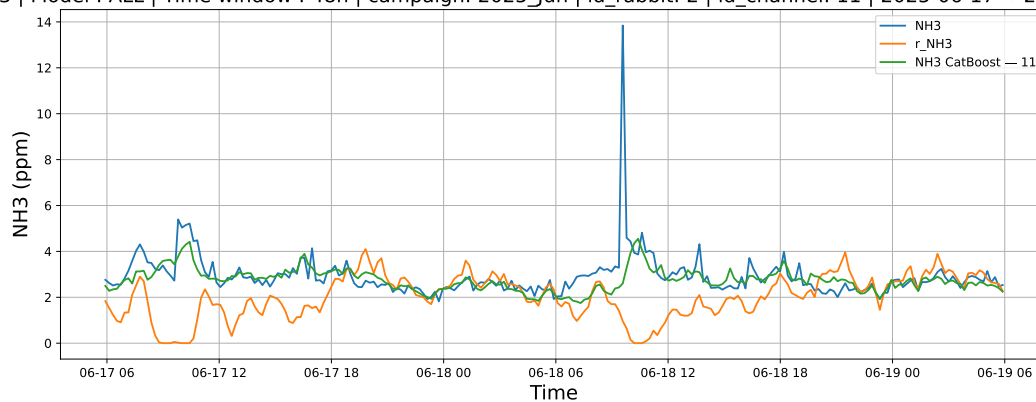
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



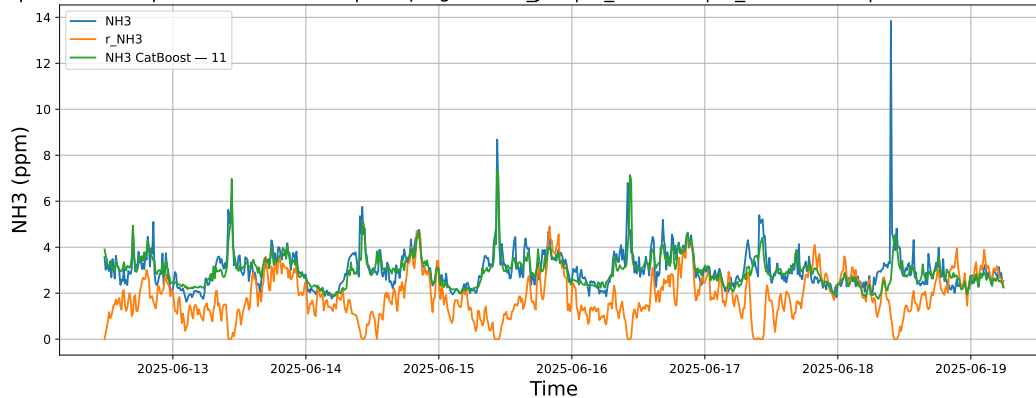
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.2.2.3 Time window : ALL

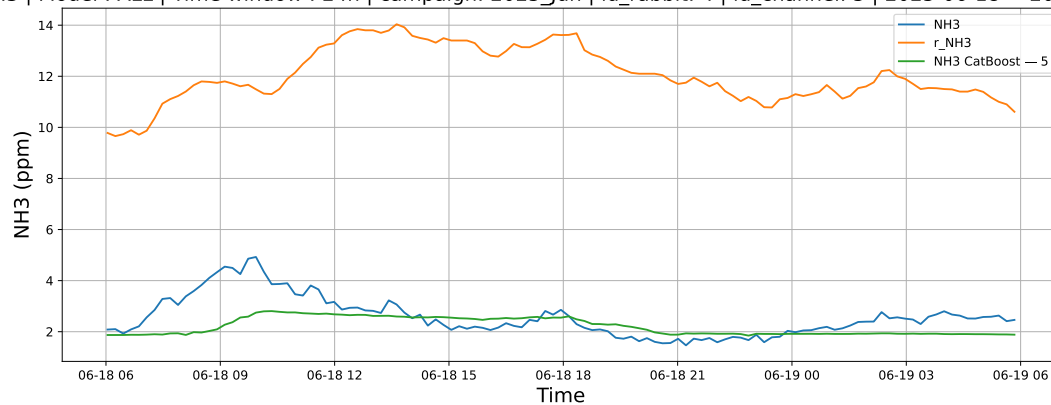
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

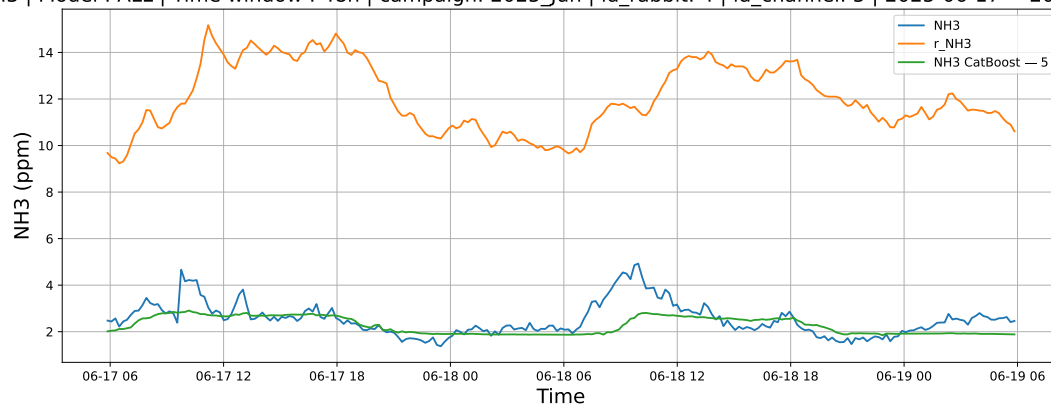
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



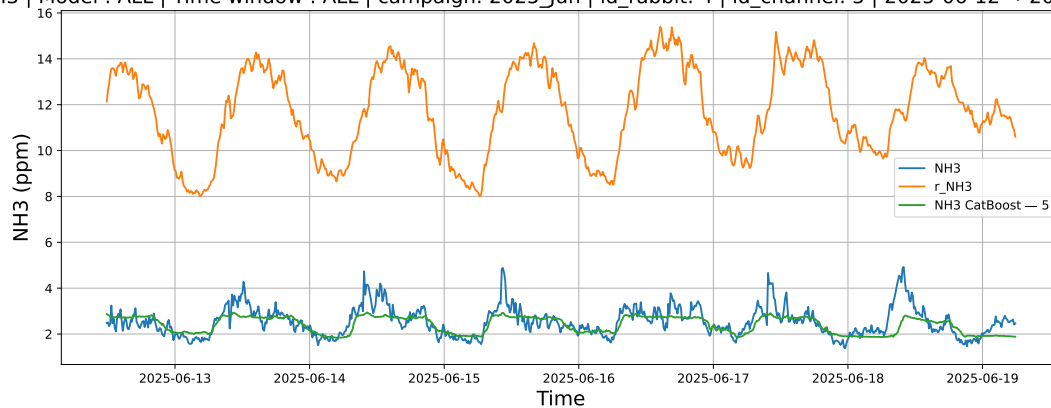
3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

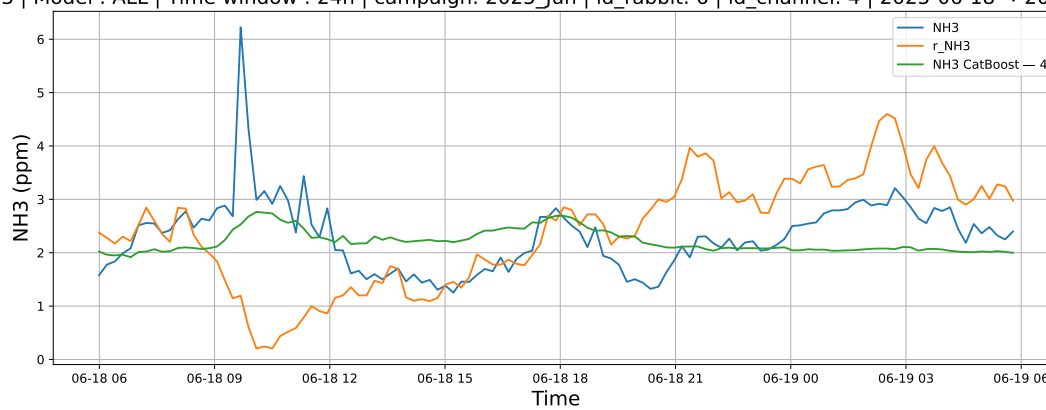
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

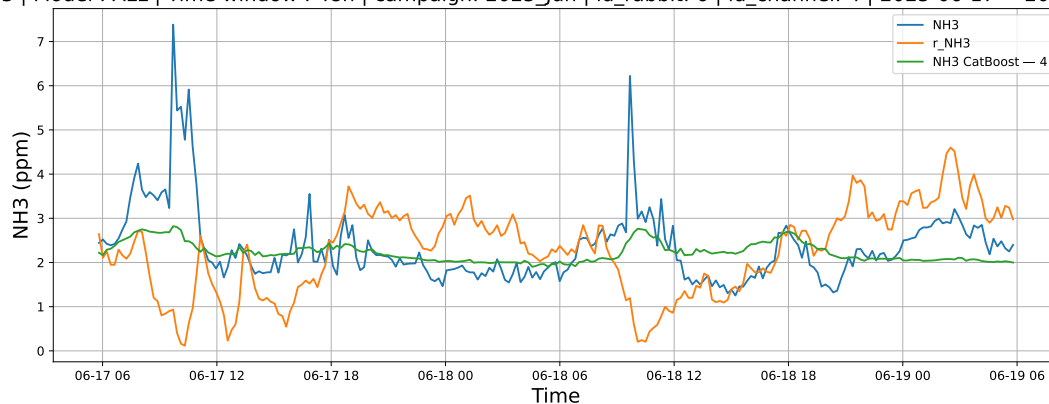
3.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



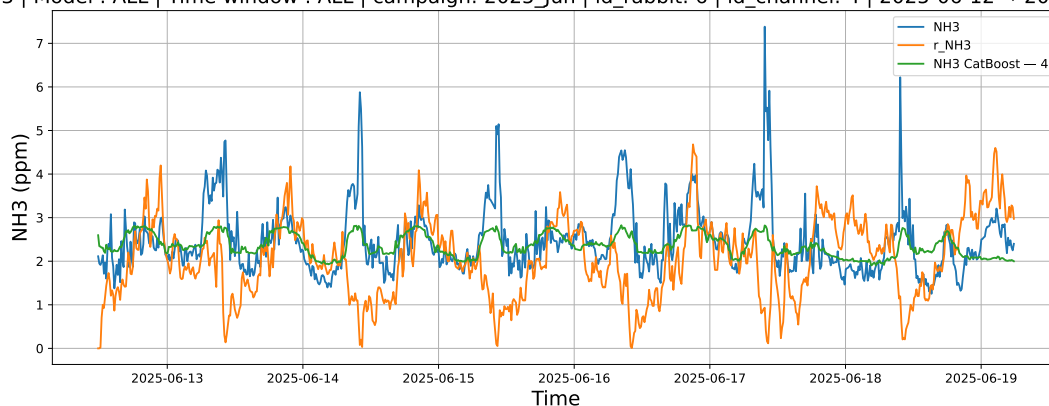
3.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.2.4.3 Time window : ALL

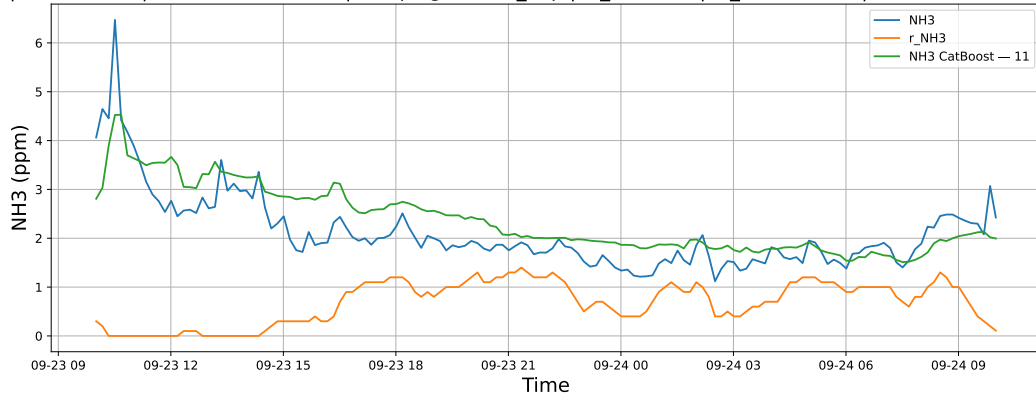
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

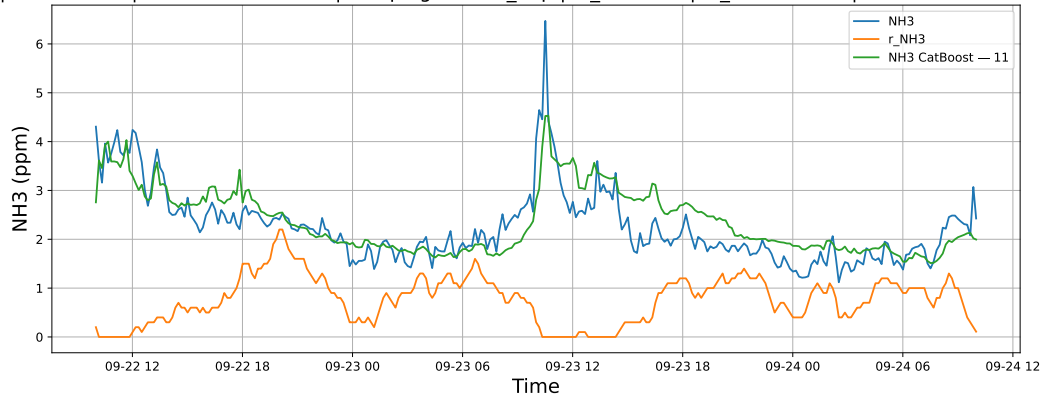
3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



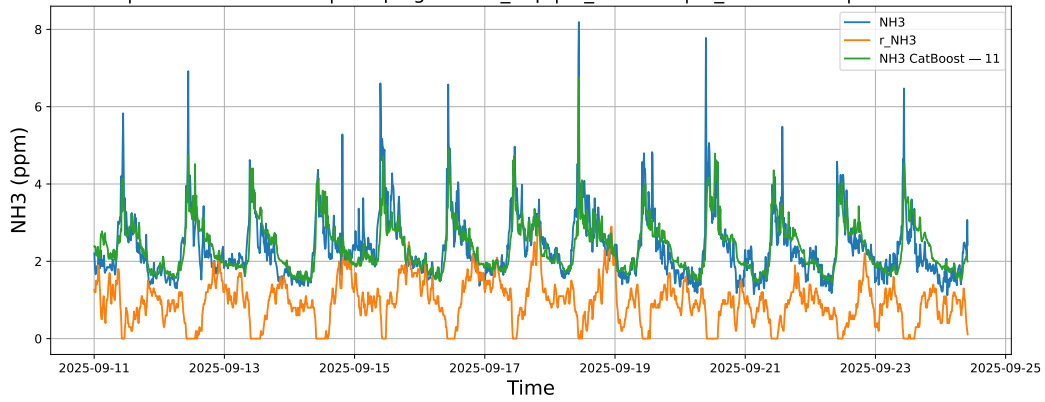
3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.2.5.3 Time window : ALL

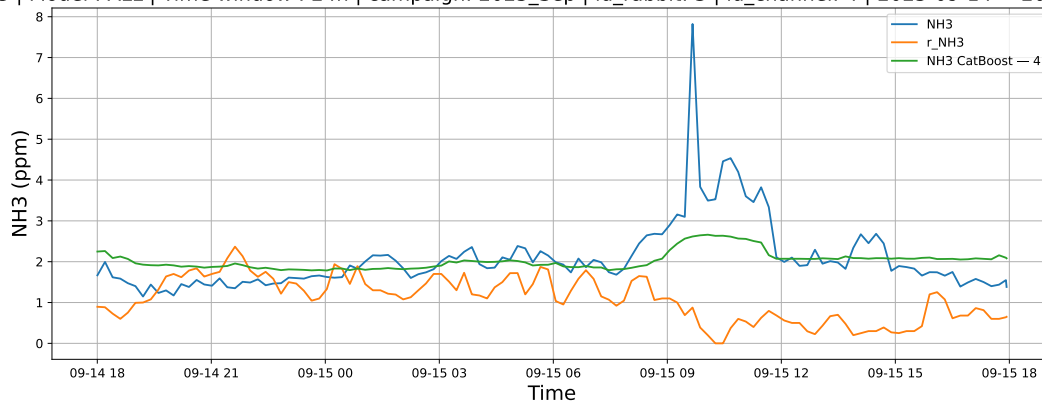
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

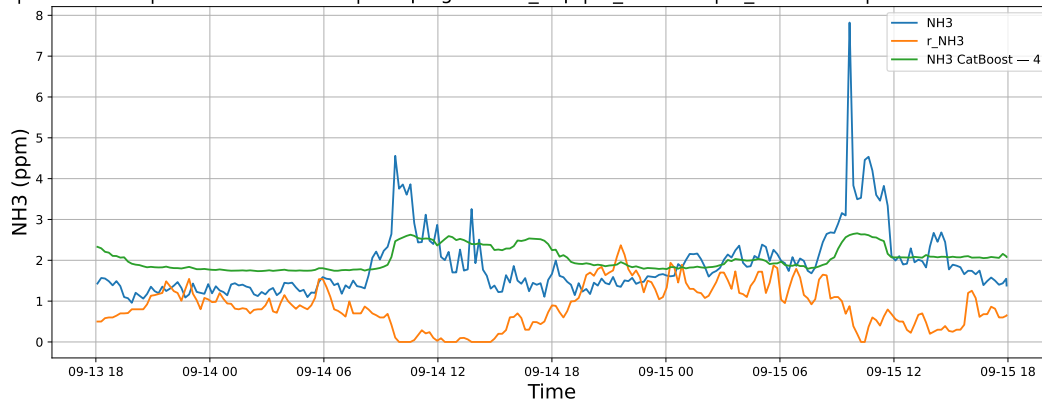
3.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



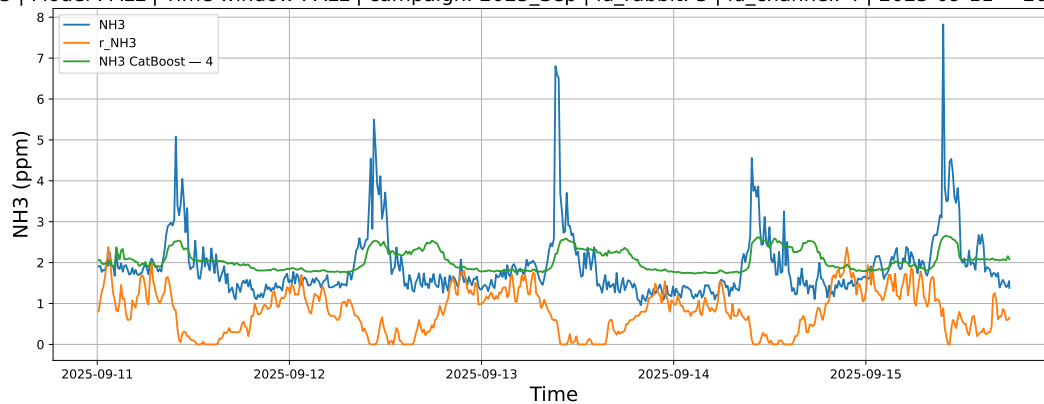
3.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.2.6.3 Time window : ALL

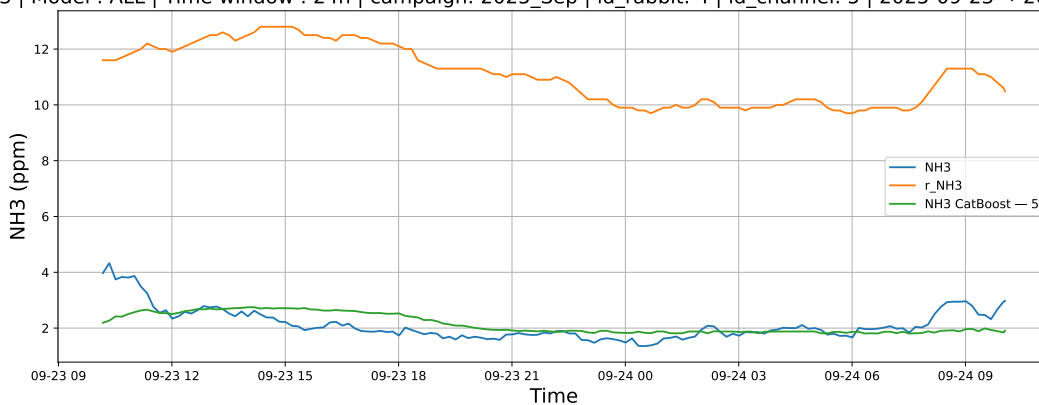
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

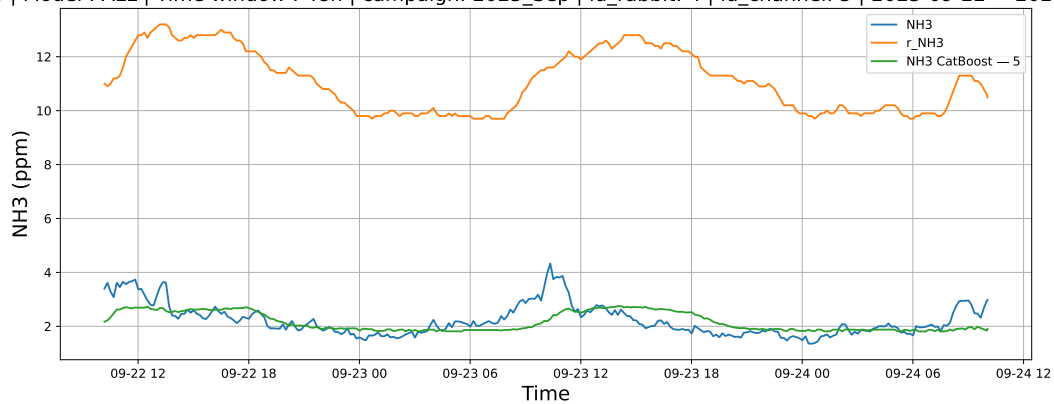
3.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



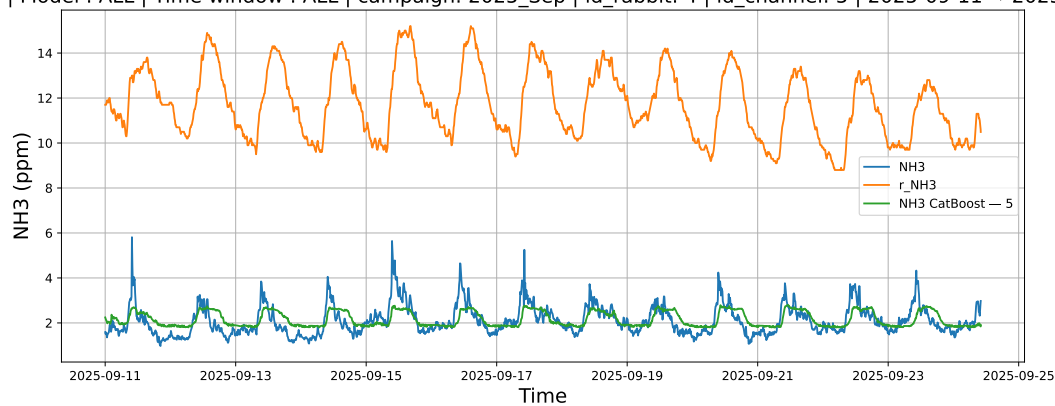
3.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.2.7.3 Time window : ALL

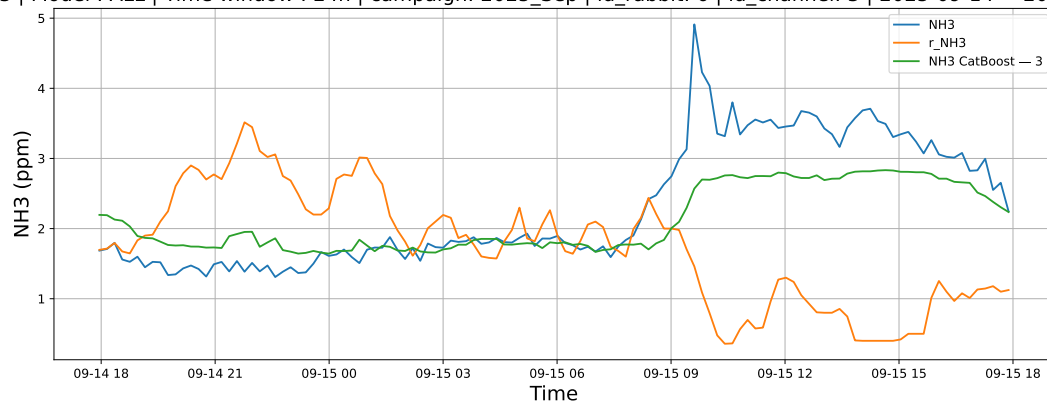
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

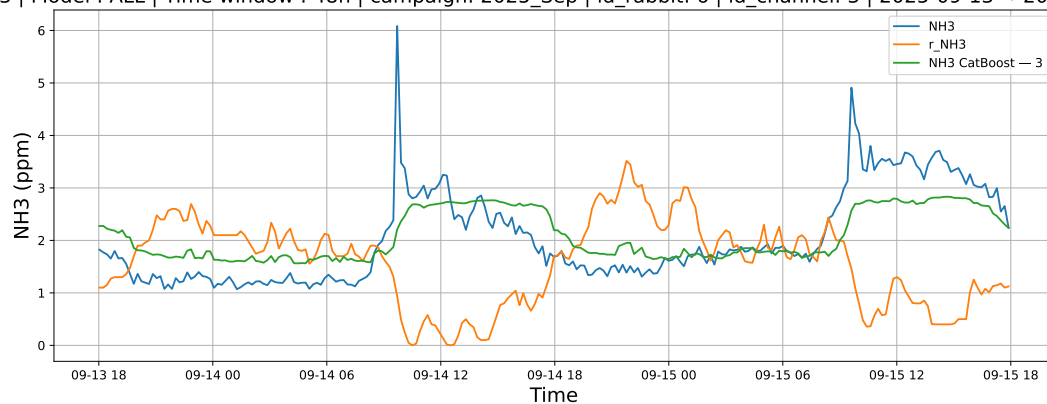
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

